

Software-as-a-Graph: Learning Cascade-Impact Rankings from Dependency Graphs of Publish–Subscribe Systems

Ibrahim Onuralp Yigit^{a,*}, Feza Buzluca^a

^a*Department of Computer Engineering, Istanbul Technical University, Istanbul, 34469, Turkey*

Abstract

Publish–subscribe middleware is fundamental to modern distributed systems, but its decoupled topic abstractions frequently obscure cascading failure paths. Software-as-a-Graph (SaG) models event-driven architectures as typed multigraphs and derives explicit dependency graphs from manifests. This study evaluates learned cascade-impact rankers—graph neural networks, hybrids, and surrogates—on twelve synthetic architectures under leave-one-scenario-out cross-validation, against three failure simulators, and zero-shot on five open-source system models.

Deriving dependency graphs enables graph learning: graph attention networks gain +0.08 to +0.11 over the raw multigraph, reaching Spearman $\rho = 0.748$ against reachability simulation. This is statistically indistinguishable from direct dependent counts (0.764, restating the simulator’s first propagation wave) and relies on degree features. Learning pays off where simulation is costly: on a dynamic queue-flow simulator requiring 12.7 CPU-hours to label the corpus, a gradient-boosted surrogate ranks held-out architectures above unweighted closed-form approximations (0.799 vs. 0.706; nominal $p = 0.019$), replacing hours of simulation with inference in seconds. While an analytical rate-weighted expansion reaches $\rho = 0.830$ without training, the surrogate captures non-linear queue saturation. Under multi-criteria fragmentation, hybrids retain accuracy via closed-form priors ($\rho = 0.585$ vs. ≤ 0.334). Pure learned models transfer best to unfamiliar systems ($\rho = 0.805$ vs. 0.662–0.695), primarily separating inert from active components; ranking within active components remains challenging ($\rho_{>0} \leq 0.342$). The results support an energy-aware two-tier approach: dependency counting for commit-stage triage, and learned surrogates for staging verification (~ 355.6 Wh theoretical TDP proxy).

Keywords: Dependency graphs, cascading failures, publish–subscribe, graph neural networks, graph learning, software architecture, dependability, empirical study

1. Introduction

1.1. Motivation and Problem

Large distributed systems now often use asynchronous publish-subscribe (pub-sub) middleware. Examples include ROS 2 for autonomous driving [1], Apache Kafka for enterprise event streams [2], DDS in cyber-physical systems [3], MQTT in IoT [4], and cloud-native microservices [5, 6]. Pub-sub separates producers and consumers by space, time, and synchronization [7]. Instead of direct references, components communicate via topics and brokers, and Quality-of-Service (QoS) policies set at deployment control reliability, durability, priority, and deadlines.

This decoupling also makes it harder to see how failures spread. Since publishers and subscribers are not directly connected, problems like outages, blocking, and backpressure can move through undetected paths involving brokers, shared topics, colocated hosts, and shared libraries [8, 9]. Failures usually appear in two

*Corresponding author

Email addresses: yigiti@itu.edu.tr (Ibrahim Onuralp Yigit), buzluca@itu.edu.tr (Feza Buzluca)

ways. In sequential cascades, a slow subscriber fills up a broker queue and gradually starves its publishers [10]. In simultaneous blasts, a crash in a shared library or a host outage can bring down all colocated services at once. These issues are not visible in architecture diagrams or static call graphs. The best time to handle these risks is before deployment, during design and continuous integration [11, 12], when runtime telemetry is unavailable. Therefore, architects must identify the most critical components, topics, and links using only configuration manifests.

Current practices leave a serious gap, which we call the Architecture–Code Gap. A distributed system might have no compile-time bugs in any service but still face major outages due to hidden single points of failure, unbuffered queue bottlenecks, or mismatched QoS contracts [13, 14]. Approaches like the Architecture Tradeoff Analysis Method (ATAM) depend on manual input [15]. Static code analysis inspects services in isolation [16, 17]. Chaos engineering [18] needs a working cluster, and pre-production tests like fault injection [19] and dependency tracing [20] require runnable environments. In continuous integration, heavy fault injection or chaos experiments use significant computing power and energy, so pre-deployment static analysis is a greener option if it is accurate enough [21, 22, 23]. Homogeneous centrality reduces typed topologies to untyped graphs [24, 25]. Graph neural networks can combine structural and contractual information, but only if the graph clearly shows failure paths. Software-as-a-Graph provides this representation, turning static deployment manifests into predictive insights before deployment and enabling a two-tier triage process to reduce developer alert fatigue.

1.2. The Software-as-a-Graph (SaG) Approach

Software-as-a-Graph (SaG) is a static analysis framework used before deployment for event-driven architectures (see Figure 1). First, it models the architecture as a typed, directed multigraph that includes Applications, Brokers, Topics, Execution Hosts, and Shared Libraries (Section 3.1). Next, it uses publish-subscribe rules to create a clear `DEPENDS_ON` dependency graph (Section 3.3). Then, it tests learned cascade-impact rankers—such as graph neural networks on both the raw multigraph and the dependency graph, hybrids, and learned surrogates—against a baseline that does not use training (Sections 4 and 5.2). The predictors only use the analysis graph. Three simulators provide ground truth by running on the base structural topology (Section 4.3). To measure architectural risk without relying on single or unproven assumptions, SaG uses a tri-oracle evaluation suite. This suite covers structural reachability (I^*), dynamic discrete-event queue saturation (I_{dyn}), and multi-criteria service fragmentation (I_{comp}). The main reachability oracle I^* works directly on communication topologies defined in the manifest, so its first-order propagation matches direct dependency counts (Section 4.4). To avoid mixing up structural alignment regarding predictive learning, the method tracks dependency counts as baseline references and carefully tests how well learned models generalize to dynamic queue flows and structural fragmentation. To bound, though not remove, this circularity, the evaluation uses three simulators with different mechanisms, reports partial correlations after removing the primary oracle, and times the cost of running I^* directly (Section 6.4). Verifying these results against real-world failures is still an important goal (Section 7.3). The conclusions and guidance here are valid as long as the simulations are accurate.

Findings in brief. Making dependencies explicit is what enables graph learning in this setting. On the raw multigraph, every relation points away from Applications, so message passing never reaches the scored nodes. Accordingly, the registered primary contrast, a heterogeneous graph transformer against the training-free baseline on that graph, showed no significant difference ($\Delta\rho = +0.069$, $p = 0.266$). On the derived dependency graph, each Application receives messages from its dependents, and the same learners improve by $+0.08$ to $+0.11$. There, a graph attention network reaches Spearman $\rho = 0.748$ against the reachability oracle. This matches the number of direct dependents (0.764) within statistical resolution. That count is publish–subscribe afferent coupling, which corresponds to the oracle’s first propagation wave. Ablations explain the match: the learner draws on its degree features and loses 0.14 when they are removed. The practical implication is a training-free triage signal for reachability impact that costs one to two orders of magnitude less than a single simulation pass. Learning adds most where simulation is expensive. On a discrete-event queue-flow oracle that takes 12.7 CPU-hours to label the corpus, a gradient-boosted surrogate combining dependency, rate, payload and QoS features ranks held-out architectures at $\rho = 0.799$. It outperforms unweighted closed-form approximations (0.706; nominal $p = 0.019$) and replaces hours of simulation with inference in seconds. On

this oracle, declared QoS contracts contribute measurable signal (+0.095). When publication rates are directly incorporated into an analytical expansion ($\hat{\rho}_{\text{dyn},1}^{\text{rate}}$), it achieves $\rho = 0.830$ without training, revealing that the gain stems from message rate transport physics rather than machine learning inductive bias alone; the surrogate remains useful when complex buffer saturation must be learned. On a multi-criteria oracle that combines fragmentation and throughput loss, hybrid engines that correct a closed-form prior remain accurate ($\rho = 0.585$), whereas learners without a prior degrade ($\rho \leq 0.334$). As expected, training-free scores aligned with that oracle’s terms rank higher still. Pure learned engines, in turn, transfer best zero-shot to five stylized models of open-source systems ($\rho \approx 0.81$ vs. 0.66–0.70 for hybrids). They do so mainly by separating components that propagate failures from those that do not, with transitive-dependent counts ranking even higher. Ordering among active components remains the open challenge ($\rho_{>0} \leq 0.342$). These results translate into concrete guidance. Recovering 80% of the true critical set requires reviewing roughly the top 40% of Applications, so rankings are best used to prioritize architectural review rather than to block commits. Feature extraction costs 4.5–72.5× one reachability pass, so dependency counting suits commit-stage triage, while learned surrogates suit staging, where they replace expensive simulation (Section 7.1). All findings are relative to simulated failures.

1.3. Research Questions

- **RQ1 (Ranking accuracy):** *How accurately do learned engines rank components by cascading-failure impact on unseen architectures, against a training-free baseline, and how close do they come to reference rankings that restate the oracle’s propagation rule?*
- **RQ2 (What learning needs):** *What do learned engines need: which graph they read, their degree features, relation-specific parameters, or QoS inputs, once model capacity and edge-channel width are matched?*
- **RQ3 (Transfer):** *How well do learned engines trained using synthetic architectures transfer zero-shot to stylized architectural models hand-authored by a single modeler based on five open-source systems, and does within-cascade discrimination hold on the active stratum?*
- **RQ4 (Cost and Sustainability):** *What does learned ranking cost at CI/CD time with respect to latency and computational energy, which stage dominates, and how does it compare with running the simulation directly?*

The evaluation uses a version-controlled analysis plan. Only two contrasts in the plan are confirmatory. We registered the matched 2×2 , the hybrids, the reference counts, the dependency-graph learners, the full-population queue-flow labels, and the sensitivity arms as planned extensions. We registered each before its own tests ran but after we knew the main result, and we report them as secondary results. The rest of the analysis is exploratory (Section 5.3). One change from the plan, an unused selection rule, is disclosed and tested. The replication repository records the revision history and shows how these four research questions fit into the plan.

1.4. Contributions

This paper is evaluated using leave-one-scenario-out (LOSO) cross-validation on twelve synthetic architectures, against three failure simulators, and zero-shot on five hand-crafted models of open-source systems. It contributes:

1. **Explicit dependency derivation for publish–subscribe architectures** (Section 3). Typed rules turn deployment manifests into a dependency graph that exposes sequential cascades through shared topics and simultaneous blasts through shared libraries. This representation is what makes graph learning effective (+0.08 to +0.11 over the raw multigraph), and its library-mediated rule adds +0.058 in transitive reach. Infrastructure and colocation rules are defined for completeness.
2. **A reference criterion for oracle circularity** (Section 4.4). We distinguish rankings that restate a simulator’s own computation, which measure how circular the oracle is, from rankings that predict it. Applied here, the criterion shows that publish–subscribe afferent coupling recovers most of the reachability oracle. It gives a general way to interpret learned models evaluated against simulation labels.

3. **A controlled characterization of when graph learning helps** (Sections 6.1–6.3). Using matched capacity and targeted ablations, we isolate four factors: the graph the learner reads, which matters most; its reliance on degree features (-0.136 without them); relation typing, which has no effect at matched capacity; and declared QoS inputs, which add signal on the queue-flow oracle but not on reachability. We also show that anchoring learners to a closed-form prior preserves accuracy on familiar architectures and under multi-criteria failure (0.585 vs. ≤ 0.334), at a cost in zero-shot transfer (0.66 – 0.70 vs. 0.76 – 0.81).
4. **A learned surrogate for expensive dynamic simulation** (Section 6.1). A gradient-boosted model combining dependency, rate, payload and QoS features ranks held-out architectures at $\rho = 0.799$ on the queue-flow oracle, outperforming unweighted closed-form approximations ($+0.094$, nominal $p = 0.019$) and replacing an estimated 12.7 CPU-hours of simulation with inference in seconds. While an analytical rate-weighted expansion reaches $\rho = 0.830$ without training, the surrogate learns non-linear saturation when closed forms are unavailable.
5. **A reproducible multi-oracle benchmark, cost analysis, and practitioner guidance** (Sections 5, 6.4 and 7.1). The benchmark comprises seventeen architectures with 2,812 components, and three simulators label all 1,321 synthetic Applications. It is released with a version-controlled analysis plan with fourteen disclosed amendments, byte-identically regenerable datasets, and 1,548 figures mechanically reconciled against released artifacts. Latencies for counting, simulation, feature extraction and inference, measured on the same graphs with estimated theoretical energy proxies (28.0W TDP), support an energy-aware two-tier triage process: dependency counting at commit time and learned surrogates at staging, conditional on simulation fidelity.

A previous conference paper [26] introduced the publish–subscribe multigraph, the Application-level subscriber-to-publisher dependency (Rule 1 of Table 2), and a closed-form betweenness–articulation score similar to the training-free baseline (Eq. (5), unweighted and with weights $0.7/0.3$), which it validated against a reachability-loss simulation on synthetic topologies and two ROS 2 benchmarks. This paper adds the Library and infrastructure rules, QoS weighting and the QoS edge encoding, the reference criterion, the learned, hybrid and surrogate engines, LOSO and zero-shot evaluation, the second and third oracles, the matched control, and the cost analysis.

Section 2 reviews related work. Sections 3 and 4 present the model and the rankers. Sections 5 and 6 cover the evaluation. Section 7 discusses the findings and threats, and Section 8 concludes.

2. Related Work

2.1. Dependability Analysis of Distributed Systems

Runtime dependability methods like broker clustering, backpressure, autoscaling, failover, and chaos engineering [18] require a running cluster, can interrupt service, and consume significant computing resources. This resource use is a major concern in green software engineering and Green AI [21, 22, 23], where the carbon footprint of large execution clusters in continuous integration encourages lightweight, energy-efficient static alternatives. Pre-production methods, such as service-level fault injection testing [19] and microservice dependency localization with PageRank (MicroRank [20]), offer lighter options before deployment. Benchmark platforms with cataloged, replayable faults, like Train-Ticket [27], help measure the impact of failures on a running deployment, which this study has not yet validated (Section 7.4). Architecture-based reliability prediction has a long history. Yacoub and Ammar’s method for architecture-level reliability risk analysis [28] was one of the first to combine component dependency graphs with complexity and failure severity to rank components by risk. State, path, and additive reliability models reviewed by Goseva-Popstojanova and Trivedi [29] are classic examples, as is the Palladio Component Model [30]. These methods answer broader questions but require operational profiles and failure rates unavailable at commit time. SaG focuses on a narrower question using only manifests: whose failure has the widest impact in the declared topology?

Telemetry-based methods find faults in running microservices. Examples include Seer [31] and MicroRCA [32], which Zhang et al. [33] thoroughly review. In synchronous microservices, failures can spread because of thread-pool exhaustion, RPC timeouts, and retry storms [27]. In publish-subscribe systems, issues spread

through queue saturation and message starvation. All these methods need data from a live system, but SaG works before any code is run. SaG’s focus is similar to change impact analysis [34] and dependency management in microservice fleets [35]. Ranking by impact is related to learning-to-rank defect prediction [36], which explicitly optimizes the ranking measure, as our listwise loss does (Section 4.2).

2.2. Static Code Analysis and Static System Analysis

Static code analysis (SCA) tools like SonarQube [16] measure complexity [37], cohesion, and coupling [17] within individual services to identify modules likely to have defects [38, 39]. Software engineering defect prediction has long compared advanced graph measures with simple structural metrics. For example, Zimmermann and Nagappan [40] studied network analysis on dependency graphs, while Premraj and Herzig’s replication study [41] showed that simple code and coupling metrics frequently perform as well as network measures. SCA cannot detect inter-service messaging, broker saturation, or issues that spread across hosts. Architecture recovery tools rebuild system-level structure from code [42], and HAROS [43] checks ROS systems before launch. SaG’s static system analysis (SSA) uses the declared topology instead, spreading code-level metrics across architectural dependencies. Architecture-level coupling metrics count a service’s consumers, such as afferent coupling [44], the Absolute Importance of the Service [45], and service fan-in [46]. In publish–subscribe systems, these counts are not simply a single edge lookup, since consumers reach producers only via topics, brokers, and shared libraries. They require a typed two-hop query (publisher to topic to subscriber), which SaG’s dependency rules define clearly. Because this count corresponds to the first propagation wave of the reachability oracle used here, this paper reports it as a reference that restates the oracle, not as a predictor (Section 4.4). This helps teams find structural anti-patterns [47, 48] and architectural technical debt [49] in CI/CD [50].

2.3. Multi-Criteria Weighting

Combining QoS policies or failure effects into a single, auditable weight is a multi-criteria decision problem. The Analytic Hierarchy Process (AHP) [51] is the standard method, and SaG uses it for the topic weights in Section 3.2 and the composite oracle in Section 4.3. AHP’s consistency ratio can find inconsistent judgments, but it does not detect matrices that have been filled in to match a chosen answer. This study notes that distinction for its weights.

2.4. Graph Learning and Explainability

Closed-form indices [24, 25, 52, 53], including one used as this study’s training-free baseline, and cascade models of network robustness [10, 8, 9] assume homogeneous, usually undirected graphs. This means a single untyped score can mix up different elements like topics, libraries, and hosts. Learned node-importance methods, such as FINDER [54], DrBC [55], and PowerGraph [56], also assume homogeneity, as does learned approximation of betweenness and closeness [57]. Homogeneous GNNs like Graph Attention Networks (GAT [58]) ignore relation identity unless it is added as a feature. Softmax attention averages over neighbors and cannot count them, while sum aggregation can [59, 60]. Counting larger substructures is still hard for standard message-passing architectures. Since the main reference here is neighbor count, this difference matters (Section 6.2). Heterogeneous GNNs, like the Heterogeneous Graph Transformer (HGT [61]), learn transformations specific to each relation. We test both types with the same capacity (Section 6.2). Khodabandeh et al. [62] use graph attention on microservice call graphs to predict upcoming interactions. In contrast, we predict the effect of removing a node from a declared topology. GNN explainers like GNNExplainer [63] describe models using their internal features. SaG’s explanation layer instead names ISO/IEC 25010 quality sub-characteristics [64] and a remediation class for each flagged component, but this is not evaluated here.

This study aligns with research showing that simple baselines perform as well as more advanced learning methods [65, 66]. In software engineering, it also supports findings that network measures on dependency graphs [40] add little value over simple coupling metrics once those are included [41]. The main contribution here is an evaluation of learning on derived dependency graphs for a pre-deployment ranking task. The results are mostly negative when the oracle is inexpensive, and its best simple version is a known architecture metric, but positive for learned surrogates when the oracle is costly.

3. The Software-as-a-Graph (SaG) Architectural Model

Figure 1 shows the SaG framework’s complete architecture. The shared front end handles Architecture-as-Code manifests, builds a typed multigraph, maps out runtime interactions in a logical dependency layer, and extracts typed node properties. The ranking engines (Section 4) use these features and, separately (with no shared parameters), the proposed explanation layer does as well.

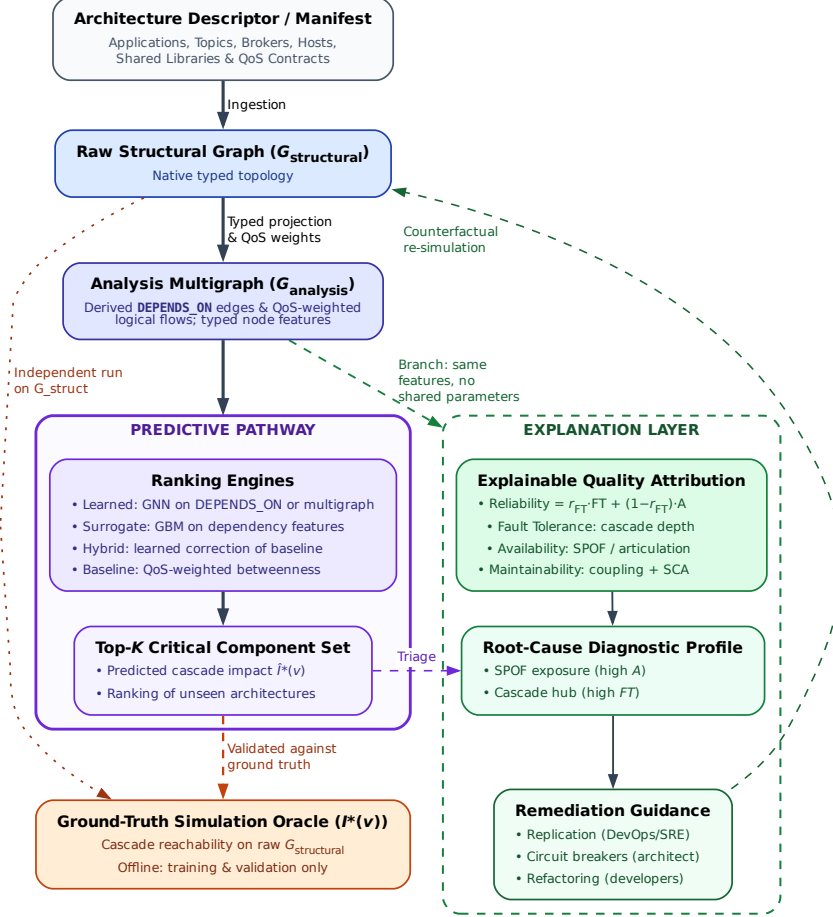


Figure 1: End-to-end architecture of the SaG framework. The predictive pathway runs down the center: manifest ingestion, typed multigraph, **DEPENDS_ON** projection with typed node properties, the ranking engines (training-free baseline, learned and hybrid; Figure 3), and the ranked critical set. The dashed edge marks the ground-truth simulation oracles, which operate only on $G_{\text{structural}}$, train the predictor offline, and do not participate in inference. The proposed explanation layer (not evaluated) reads the same analysis multigraph, shares no parameters with the predictor, and applies to components after they have been ranked; no predictor output flows into it.

3.1. Multigraph Definition

A distributed system is described as a typed, weighted, directed multigraph:

$$\mathcal{G} = (V, E, \tau_V, \tau_E, w_V, w_E) \quad (1)$$

where:

- $V = V_{\text{app}} \cup V_{\text{broker}} \cup V_{\text{topic}} \cup V_{\text{host}} \cup V_{\text{lib}}$ holds the five entity types $\mathcal{T}_V$ of Table 1; V_{host} denotes physical or virtual *Execution Hosts*.
- E is the set of directed edges, and $\tau_V : V \rightarrow \mathcal{T}_V$ and $\tau_E : E \rightarrow \mathcal{T}_E$ assign entity and relation types.

- $w_V : V \rightarrow (0, 1]$ and $w_E : E \rightarrow (0, 1]$ weight entity criticality along with connection strength. For Applications and Libraries, $w_V(v) = 1 - \text{CQP}(v)$, where CQP is a code-quality penalty computed from static code metrics (lines of code, cyclomatic complexity, coupling and cohesion); otherwise $w_V(v) = 1.0$.

The raw multigraph is denoted $G_{\text{structural}}$ (used exclusively by simulation oracles), while G_{analysis} denotes the logical **DEPENDS_ON** projection consumed by predictors. A comprehensive reference of mathematical notation is provided in the replication repository.

Scope of Evaluation and Entity Roles. While the multigraph formalizes five entity types, this study concentrates its empirical study specifically on *Applications* (V_{app}). In this work, the term *Application* is adopted to align with publish-subscribe standards (such as OMG DDS and ROS 2 participant nodes); conceptually, it corresponds directly to individual microservice containers, autonomous service agents, or communicating daemon processes in distributed architectures. Applications represent the developer-authored, code-level services that are often changed, refactored, and tested in CI/CD pipelines, so they are the main focus for pre-deployment triage. Brokers, Topics, Execution Hosts, and Libraries are also modeled and included because they are important for understanding indirect failure propagation. For example, we cannot analyze message routing, shared fault-domain colocation, or library-related failures without them. However, ranking infrastructure components like Brokers and Hosts is more relevant to operational provisioning than to software triage at commit time, so we treat this as a separate operational issue.

Table 1: Entity types and structural edge types in the SaG model.

Entity Type ($\mathcal{T}_V$)	Architectural Role	Concrete System Examples
Application (V_{app})	Process producing/consuming messages	ROS 2 node, Kafka microservice, MQTT client
Broker (V_{broker})	Message routing and queuing intermediary	RabbitMQ exchange, Mosquitto, EMQX broker
Topic (V_{topic})	Named logical communication channel	<code>/sensor/lidar</code> , <code>orders.payment.completed</code>
Execution Host (V_{host})	Physical or virtualized execution environment	Bare-metal server, Kubernetes worker, Cloud VM
Library (V_{lib})	Shared software package or runtime dependency	Kafka client, OpenCV, Protobuf runtime
Structural Edge ($\mathcal{T}_E$)	Direction	Semantic Meaning
PUBLISHES_TO / SUBSCRIBES_TO	App/Library $\rightarrow$ Topic	Component publishes to / consumes from topic
ROUTES	Broker $\rightarrow$ Topic	Broker manages and routes topic traffic
RUNS_ON	App/Broker $\rightarrow$ Host	Process is hosted on host
CONNECTS_TO	Host $\rightarrow$ Host	Network link between hosts
USES	App $\rightarrow$ Library	Application links to shared library

3.2. Quality-of-Service Link Weighting

A link’s strength is determined by its Quality-of-Service (QoS) contract. For example, a **RELIABLE** topic with **TRANSIENT_LOCAL** durability creates a stronger connection between services than a **BEST_EFFORT** telemetry stream. Each topic t has an overall weight $w(t) \in (0, 1]$ that combines its QoS policies (reliability, durability, priority) with payload size and publication frequency. We set the reliability, durability, and priority weights employing an Analytic Hierarchy Process (AHP) pairwise-comparison matrix, which we defined independently and did not adjust to fit a target vector ($CR = 0.016$, non-degenerate; the framework’s other AHP matrices also use a declared vector).

We test the effect of these weights directly on a learned ranker. When both cases include the weighted in-degree, the QoS inputs do not improve the graph neural networks on the reachability oracle (Section 6.2). However, for the queue-flow oracle, they do benefit the learned surrogate (Section 6.1). Details on how weighting affects the training-free baseline are available in the replication repository. The dependency counts reported in this paper are unweighted and do not use $w(t)$.

3.3. Logical Dependency Projection (**DEPENDS_ON**)

Structural edges do not directly show how failures spread. For example, a subscriber depends on a publisher, but pub-sub topologies have no direct edge between them. SaG creates an explicit semantic relation called **DEPENDS_ON**, which goes from the dependent to the dependency. This means if the target fails, the source is affected. Table 2 lists the rules.

Table 2: The DEPENDS_ON logical dependency projection rules. [†]Defined for completeness; not exercised by the Application-level evaluation.

Rule	Dependency Category	Structural Pattern (Dependent $\rightarrow$ Dependency)	Derived Weight (w)
1	app_to_app	Subscriber $\rightarrow$ Publisher (via shared Topic)	$1 - \prod_{t \in T} (1 - w(t))$
2 [†]	app_to_broker	Publisher/Subscriber $\rightarrow$ Broker routing its topics	$1 - \prod_{t \in T} (1 - w(t))$
3 [†]	host_to_host	Host $\rightarrow$ Host (lifted from inter-host app dependencies)	$\max_{u \in \text{hosted}(h_1), v \in \text{hosted}(h_2)} w_E(u \rightarrow v)$
4 [†]	host_to_broker	Host $\rightarrow$ Broker (lifted from hosted app dependencies)	$\max_{u \in \text{hosted}(h)} w_E(u \rightarrow b)$
5	app_to_lib	Application $\rightarrow$ Shared Library it USES	$H(w_V(\text{app}), w_V(\text{lib}))$
6 [†]	broker_to_broker	Broker $\leftrightarrow$ Broker (shared fault-domain colocation, symmetric)	$w_V(\text{host})$

Rules 1 and 2 combine topics T that connect a pair using probabilistic union [67], so that having multiple failure paths increases the connection strength. Rule 5 uses the harmonic mean $H(x, y) = 2xy/(x + y)$ [68]. Rules 3 and 4 assign dependencies to hosts by taking the maximum value.

Sequential cascades and simultaneous blasts. Rule 1 describes sequential cascades, where if a publisher fails, its subscribers are affected because they do not receive messages. Rule 5 describes simultaneous blasts, where if a library crashes, all applications that depend on it fail at the same time. Libraries that publish or subscribe are also endpoints for Rule 1, and an Application can reach a library’s topics through Rule 5. For counting, Topo-QoS and all -P learners use Rule 1 only for direct subscriptions. The repository’s method for creating node properties also follows up to three USES links, so if an Application subscribes through a library it uses, it is considered a Rule 1 dependent. Rules 2, 3, 4, and 6 cover infrastructure and broker dependencies (see Table 2), which define physical host failure domains and broker routing bottlenecks. Since this evaluation focuses on Application-level code, these four rules are not used in the ranking benchmarks but are included for completeness.

Remark 1 (the dependency count is a typed two-hop count). Let G_{flow} be the Application–Library projection (Rules 1 and 5) and v an Application. By construction,

$$\text{InDeg}(v) = |\{u \neq v : \exists t \in V_{\text{topic}}, (v, t) \in \text{PUBLISHES_TO} \wedge (u, t) \in \text{SUBSCRIBES_TO}\}|. \quad (2)$$

Since Rule 5 edges end at Libraries, every edge into an Application is a Rule 1 edge $u \rightarrow v$, which exists exactly when $u \neq v$ (an Application or a Library) directly subscribes to a topic that v publishes; parallel topics collapse into one edge.

The right side of Eq. (2) represents publish–subscribe afferent coupling (fan-in, or the Absolute Importance of the Service [44, 45]), and is a typed two-hop query on the raw multigraph. InDeg can be computed directly without projection; the projection defines which typed query to use. We checked this equality on all seventeen corpus graphs, with a maximum absolute difference of 0. We measure the extra value it adds separately: Rule 5 increases transitive reach by +0.058 (Section 6.1). Since the primary oracle’s first propagation wave matches this set (Section 4.4), InDeg and its transitive version Reach are used as reference points in this paper, not as predictors.

3.4. Dual Graph Views

The **structural graph** $G_{\text{structural}}$ is the raw deployment topology. The **analysis graph** G_{analysis} adds the DEPENDS_ON edges and code metrics (see Figure 2). Predictor features are computed on G_{analysis} , while simulation oracles run strictly on $G_{\text{structural}}$ (Section 4.4). The two views also differ for a learner: on $G_{\text{structural}}$, every relation points away from Applications, so forward message passing does not reach them. In the DEPENDS_ON graph, each Application receives messages from its dependents (Section 6.2).

3.5. Typed Node Feature Encoding

Both the predictive pathway (Section 4) and the proposed explanation layer use typed node properties from G_{analysis} . All five entity types share a base set of 18 normalized topological metrics (ranging from $[0, 1]$): PageRank, Reverse PageRank, betweenness, closeness, eigenvector centrality, in-degree, out-degree, clustering, articulation, bridge ratio, node QoS weight, incoming and outgoing dependency weights, multi-path coupling, path complexity, fan-out criticality, and the Connectivity Degradation Index (CDI). We calculate four centrality features (PageRank, Reverse PageRank, betweenness, and closeness) over weighted edges, so they

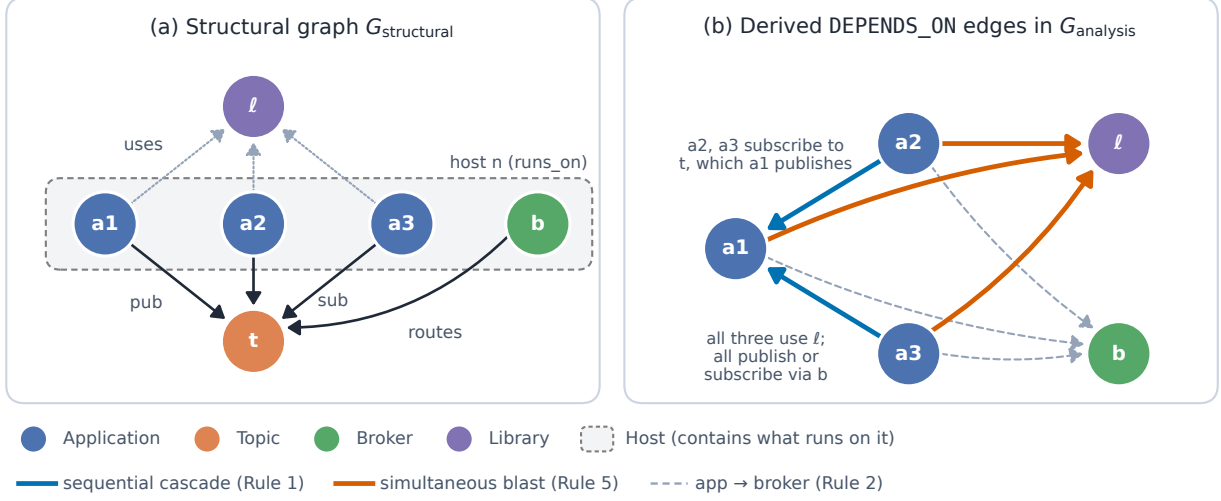


Figure 2: Running example. (a) Three applications share topic t (routed by broker b) and library l , and all run on host n . No structural edge joins two applications. (b) The derived DEPENDS_ON edges make the hidden dependencies explicit: subscribers a_2, a_3 depend on publisher a_1 (Rule 1), all applications depend on l (Rule 5), and each application depends on the broker (Rule 2). Simulation oracles run on view (a); predictors read view (b).

include QoS information. This means the so-called QoS-off condition is not entirely free of QoS effects. The in-degree feature and the incoming dependency weight w_{in} (a QoS-weighted version) ensure that all learners receive information closely related to InDeg , so the learners inherit some overlap with the oracle’s reference counts (Section 6.2). We did not use any learner missing both columns. Type-specific blocks add more features, extending the vector to 19–25 dimensions (the full schema is in the replication repository). CDI measures structural connectivity loss using a fixed-scale breadth-first sample to keep computation time reasonable (Section 6.4). We discuss its sensitivity to hash-seeded tie-breaking in Section 7.3.

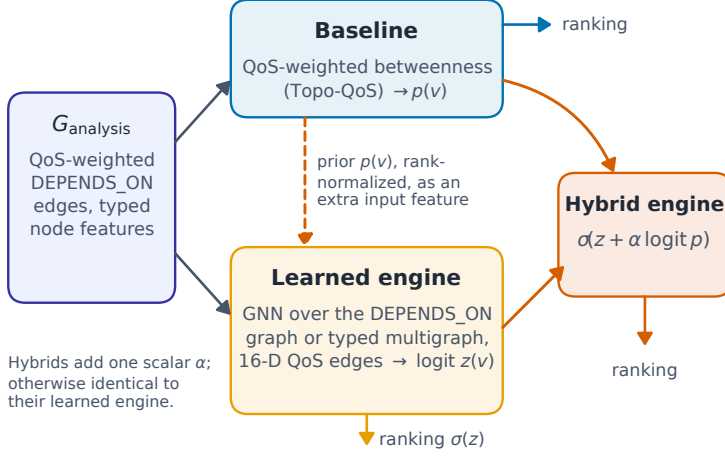
4. Ranking Engines and Ground Truth

SaG uses learned engines to rank components and compares them to a training-free baseline. The learned engines are graph neural networks built on the typed multigraph and the dependency graph, as well as gradient-boosted models using per-Application dependency-graph features (Section 5.2). Hybrid engines are graph neural networks that adjust the baseline score (Section 6.1). The training-free baseline, called **Topo-QoS**, uses QoS-weighted betweenness on the dependency projection (Section 5.2). We train or evaluate all models using simulation oracles on a separate graph view (Sections 4.3–4.4). Figure 3 illustrates the relationships between the engines and their evaluation process. Full hyperparameters and training commands are available in the experiment pages of the replication repository (Section 5.1).

4.1. Heterogeneous Graph Transformer and Attention Networks

The main learned engine, **HGT-QoS**, is a three-layer Heterogeneous Graph Transformer (HGT) [61] built with PyTorch Geometric [69]. It uses a hidden dimension of $D = 64$ and $H = 4$ heads. Entity-specific linear projections map raw node features $x_v \in \mathbb{R}^{19-25}$ into the hidden space, with $h_v^{(0)} = \text{LayerNorm}(\text{GELU}(W_{\tau(v)}x_v))$. For each meta-relation $\langle \tau(u), \phi(e), \tau(v) \rangle$, attention uses type-parameterized keys, queries, and values, scaled by a learned per-meta-relation prior $\mu_{\langle \tau(u), \phi(e), \tau(v) \rangle}$; full layer-wise formulations are in the replication repository. Message passing runs over both G_{analysis} and its transpose. In the raw multigraph, native relations point away from Applications, so forward message passing cannot reach the scored nodes. This means forward GNNs act as per-node MLPs over precomputed centralities (Section 7.2). The baseline Graph Attention Network (**GAT-QoS** and **GAT-P-QoS** for the dependency projection) uses a three-layer architecture with four attention heads and width $D = 288$ ($D = 296$ in the unweighted control for parameter matching), projecting all node types into a joint embedding space before applying homogeneous **GATConv** layers.

(a) Learned engines and baseline, one analysis graph



(b) Ground truth and evaluation

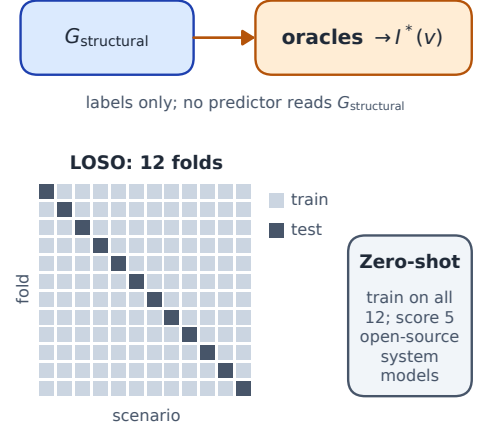


Figure 3: (a) SaG’s ranking engines read the analysis graph. The training-free baseline scores QoS-weighted betweenness; $p(v)$ is that **Topo-QoS** score, rank-normalized to $[0, 1]$ within the graph; the learned engine outputs a logit $z(v)$. A hybrid engine gives the learned engine $p(v)$ as an extra input feature and adds a learned correction to it on the logit scale, $\sigma(z + \alpha \logit p)$, with one learnable scalar α . (b) Ground truth comes from simulation oracles on the structural graph, which no predictor reads. We evaluate engines by leave-one-scenario-out cross-validation over twelve synthetic architectures and zero-shot on five open-source system models.

Each directed edge has a 16-dimensional vector. Index 0 represents the coupling weight, index 1 is the normalized path count, indices 2 to 8 are a one-hot encoding of the relation, and indices 9 to 15 are middleware QoS parameters such as reliability, durability, priority, depart-mode flag, deadline pair, and max-blocking time. The edge vector is projected and added before attention.

4.2. Prediction Head and Training Objective

A composite head predicts the simulated cascade impact, supplemented by an active auxiliary head $\hat{a}_1$ regressing on the simulator’s reliability column I_R^* . (A secondary maintainability head defined in the schema is masked out of backpropagation as ground-truth maintainability is unrecorded by the simulator.) The operational training objective combines mean-squared error regression with listwise and margin-ranking terms:

$$\mathcal{L} = \text{MSE}(\hat{I}^*, I^*) + 0.5 \cdot \text{MSE}(\hat{a}_1, I_R^*) + 0.3 \cdot \mathcal{L}_{\text{rank}} + 0.1 \cdot \mathcal{L}_{\text{pairwise}}, \quad (3)$$

where $\mathcal{L}_{\text{rank}}$ is ListMLE [70] over the ground-truth permutation π ,

$$\mathcal{L}_{\text{rank}} = -\frac{1}{N} \sum_{i=1}^N \left(\hat{s}_{\pi_i} - \log \sum_{j=i}^N \exp(\hat{s}_{\pi_j}) \right), \quad (4)$$

and $\mathcal{L}_{\text{pairwise}}$ is a margin-ranking loss with $\gamma = 0.05$ over pairs differing by more than γ . Tied labels, which affect about 31% of Applications with $I^* = 0$, enter π in a fixed but arbitrary order, so ListMLE is only approximately defined on them; the pairwise term ignores pairs among γ .

Models are trained with AdamW ($\eta = 3 \times 10^{-4}$, weight decay 10^{-4}) using cosine warm restarts for up to 300 epochs. We set early stopping at 60 epochs of patience on a 20% node-level validation split from the largest training scenario, using five fixed seeds ($\{42, 123, 456, 789, 2024\}$). We set hyperparameters and loss coefficients before the first sweep and kept them the same across all experiments. This approach differs from the registered plan, which called for selecting every hyperparameter using nested inner-LOSO selection (two-fold inner holdout). Instead, the published results use one fixed configuration, chosen during the main experiments to keep computation manageable across seeds and folds. We explain this difference in Section 5.3.

We ran the registered selection later as a sensitivity analysis on the eight-configuration stage-1 grid (varying depth and rank normalization of features and labels). This analysis also used early stopping on a held-out training scenario, which is another difference from the published protocol (Section 6.2). The results confirm that the main null finding does not change under nested tuning.

4.3. Ground-Truth Simulation Oracles

Ground truth is evaluated using simulation oracles on the raw structural multigraph $G_{\text{structural}}$:

- **Primary reachability cascade oracle (I^*):** Crashes component v , propagates outages through dependent topics, brokers and links via breadth-first traversal on $G_{\text{structural}}$, and computes the mean fractional feed loss across intact subscribers. Feed loss is scaled by a QoS severity ladder ($\times 1.2$ RELIABLE, $\times 1.15$ high priority, $\times 1.05$ medium) and clamped to $[0, 1]$. Each propagation wave is processed in a seeded random order, which decides ties at the propagation threshold; the label is the mean over five seeds. Disabling QoS scaling leaves the Application ordering largely intact ($\rho = 0.965$ across the twelve folds), and substituting durability-aware rescaling moves it less still ($\rho = 0.977$), showing that I^* is predominantly a topological reachability metric.
- **Independent queue-flow discrete-event oracle (I_{dyn}):** A discrete-event SimPy [71] simulation to model dynamic message emission rates, queue buffer saturation, and network latency. It measures the drop in delivered message rate for surviving consumers under load, using declared rates, payload sizes, and QoS contracts that I^* ignores. Its mechanism differs from reachability, but it shares the same pub-sub topology, and its first-order effect—consumers that lose a failed publisher’s messages—is again subscriber loss. While I_{dyn} models declared rates and buffer limits, it operates on configured parameters free of physical operating-system dynamics like socket buffer backpressure, thread scheduling, or garbage collection pauses. It provides labels for all 1,321 Applications, averaging over five seeds, and takes 12.7 CPU-hours for the full corpus, compared to seconds for I^* . An earlier version used only the first 30 Applications per fold in lexicographic order as a sensitivity check. I_{dyn} agrees with I^* at a mean $\rho = 0.711$ $[0.625, 0.794]$, and its published test–retest reliability is 0.74–0.97, which sets the upper bound for correlation any deterministic ranker can achieve.
- **Composite multi-criteria failure oracle (I_{comp}):** An auxiliary benchmark uses the Validate-stage failure simulator. We run it on all 1,321 Applications with QoS weighting and a primed baseline flow. Each removal is scored as a linear combination of four degradation components: $I_{\text{comp}}(v) = 0.35 \text{RL}(v) + 0.25 \text{FR}(v) + 0.25 \text{TL}(v) + 0.15 \text{FD}(v)$, representing Reachability Loss (RL), Fragmentation (FR; 0.70 structural component loss + 0.30 severity), TL is Throughput Loss, and FD is Flow Disruption. The weights are from one of three AHP matrices that encode a declared vector rather than elicited judgment, so I_{comp} serves as a declared multi-objective stress test to assess resilience to network fragmentation and throughput disruption, rather than just subscriber loss. Since fragmentation and flow disruption occur with almost every removal, $I_{\text{comp}}(v) > 0$ for nearly every Application. I_{comp} is never used as a training label.

4.4. Input–Label Independence and Construction Bounds

Features are extracted strictly from G_{analysis} , while simulation oracles execute on $G_{\text{structural}}$, and no simulation output is used as an input feature; a regression test in the replication package ensures this separation.

Procedural separation does not eliminate construct overlap, which is the main risk discussed in this paper. **Criterion.** A ranking is a *reference* for an oracle O if it is a closed-form simplification of O ’s own computation using the same inputs. In this case, it measures how much of O comes from its own rule, not predictive skill, so comparisons are not claims of superiority. For I^* , the first propagation wave corresponds to the set of InDeg counts (see Remark 1), so the references are its first-order expansion (Eq. (6)), InDeg, Reach, and their raw-graph equivalents. For I_{dyn} , no ranking truncates its computation, but its main effect is still subscriber loss, so we use the same rankings as references here too. For I_{comp} no ranking is a truncation either; its fragmentation and flow terms respond to removing high-degree and high-betweenness nodes, so Topo–QoS and raw degree are considered close to it, and their performance is interpreted with the same

caution as the references for I^* . The learned engines are not references by this definition, but they use in-degree features and are trained on I^* , so they share some overlap (Section 6.2 measures this). Two analyses limit the circularity instead of removing it. Every ranker is evaluated on all three oracles, and its agreement with I_{dyn} is also reported after removing the effect of I^* (and separately, the first-order term) (Section 6.1). However, none of these analyses replaces validation against real failures in live distributed systems, which this study does not include (Section 7.3). Therefore, all results depend on the simulations’ accuracy.

5. Experimental Setup

5.1. Corpus and Replication Package

The corpus includes 2,812 components from seventeen architectures (see Table 3). Twelve synthetic topologies make up the LOSO folds, covering areas like autonomous vehicles, financial trading, healthcare, industrial SCADA, smart-city IoT, telecom RAN, logistics, gaming, microservices, enterprise integration, and air-traffic management. Because all twelve topologies and their code metrics come from the same family, LOSO tests transfer across several parameter settings of that generator, not across unrelated industrial systems. We can regenerate each topology exactly from its configuration, and continuous integration checks this using a SHA-256 manifest.

Table 3: Evaluation corpus. The twelve synthetic topologies are the LOSO folds; we exclude the five open-source system models from all training and use them only for zero-shot transfer (Section 6.3). Counts are read from the committed topology files and verified in CI; the replication repository details per-scenario composition.

Dataset	$ V $	$ V_{\text{app}} $	Topics	Brokers	Hosts	Libs	$ E $
Synthetic evaluation scenarios (11)	2,387	1,295	588	60	194	250	10,657
Synthetic case study — ATM (1)	74	26	27	5	8	8	261
Synthetic subtotal (12 LOSO folds)	2,461	1,321	615	65	202	258	10,918
Open-source system models (5)	351	141	120	16	32	42	700
Total	2,812	1,462	735	81	234	300	11,618

The five open-source systems are hand-crafted models. Autoware.universe (ROS 2), EdgeX Foundry, Home Assistant, and two meshes based on Online Boutique and Train-Ticket were each created by a single author as typed multigraphs using public architectural documentation. We did not mechanically extract them from source code or container manifests. Because one person modeled them, bias is possible. To gauge this risk, an exploratory calibration spot-check on the two smallest systems (Home Assistant and EdgeX Foundry) showed high node entity overlap ($J \geq 0.88$), but edge concordance diverged to $J \approx 0.65\text{--}0.72$ between independent modelers, indicating that topic routing and broker channel synthesis from documentation require non-trivial modeler discretion. Some details, like brokers, QoS profiles, code metrics, and host specifications, are partly assumed. Two models differ significantly from their originals: the Online Boutique model is a 22-application pub-sub mesh with four brokers, while the original utilizes RPC and has no broker. The Train-Ticket model treats its service-discovery registry as an event broker. In both cases, we changed synchronous RPC interactions to asynchronous request-and-response topics with routing intermediaries (see Supplementary Section S.36). This preserves the direction of dataflow but changes synchronous blocking calls into decoupled event streams, leaving out client-side timeout and thread-pool exhaustion effects. Neither model includes synchronous call edges. As a result, RQ3 tests zero-shot transfer to stylized architecture models inspired by open-source systems made by a single modeler, not to real production systems or automatically extracted call graphs.

Replication package. All datasets, harnesses, checkpoints, and result artifacts are stored on Zenodo (see Data Availability). In addition, the public repository provides documentation for each experiment, including its protocol, hyperparameters, make target, artifacts, and extended results (<https://github.com/onuralpygit/software-as-a-graph/tree/main/docs/research/jss/experiments>).

5.2. Predictors

Table 4 shows the predictors: the training-free baseline (Topo-QoS), learned models on the dependency graph and raw multigraph, and hybrid models. Other training-free baselines (Degree-raw, PR-raw, RevPR-raw, Topo) are available in the replication repository. Dependency counts are not included in the table because they repeat the main oracle’s propagation rule and are reported as reference rankings below. For GNNs, ‘-QoS’ means using the 16-dimensional QoS edge vector (see Section 4.1) along with three QoS node columns (w , w_{in} , w_{out}). ‘Hybrid-X’ refers to engine X adjusted by the baseline prior. GAT and GAT-QoS are untyped GATs with the same parameter budget as HGT, and GAT-QoS uses the same 16-D edge vector as HGT-QoS. Together, these define a 2x2 grid based on typing (GAT vs. HGT) and the QoS channel. The study also tested smaller and projection-based variants, which are listed in the replication repository. The learned and hybrid models use the native typed multigraph under LOSO, while the ‘-P’ models use the Application–Library DEPENDS_ON graph, with identical graph node features and labels. In QoS-off settings, all edge weights are set to 1, and the QoS node columns are set to zero, but four centralities still use QoS (see Section 3.5).

Table 4: Predictors reported in the main text. -QoS: QoS-weighted distances (Topo-QoS) or the 16-D QoS edge vector plus three QoS node columns (GNNs); -P: trained on the derived dependency graph; Hybrid-X: engine X corrected by the Topo-QoS prior (the supplementary variant GAT-P+InDeg incorporates the InDeg reference prior). Unsuffixed GNNs (GAT, HGT) serve as raw-multigraph confound controls ($G_{\text{structural}}$). ¶Topo-QoS carries an implementation defect where the articulation point term is zero (corrected score is 0.533). Reference rankings that restate the oracle (Analytic I^* , InDeg, Reach) and GAT-P+InDeg, which uses one as its prior, are not predictors. We archive other training-free baselines and exploratory raw-graph references (Pubs-raw, Reach-R1) in the replication repository. Each dependency-graph learner keeps the architecture and parameter budget of its raw-multigraph counterpart.

Predictor	Graph read	Typing / edge input	Params
<i>Training-free baseline (registered comparator)</i>			
Topo-QoS¶	App-Lib DEPENDS_ON	scalar $w(e)$; betweenness only (AP term zero; corr. 0.533)	0
<i>Learned, dependency graph</i>			
GAT-P-QoS, HGT-P-QoS	App-Lib DEPENDS_ON	homogeneous / heterogeneous	429,992 / 434,620
<i>Learned, raw multigraph: typing $\times$ QoS channel at matched budget</i>			
GAT / GAT-QoS	$G_{\text{structural}}$ + features	homogeneous; none / 16-D	437,496 / 429,992
HGT / HGT-QoS	$G_{\text{structural}}$ + features	heterogeneous; 1-hot / 16-D	434,620
Hybrid-GAT / Hybrid-HGT	$G_{\text{structural}}$ + features	as base + Topo-QoS prior	431,433 / 434,941

Training-free baseline. Topo-QoS is computed on the Application–Library DEPENDS_ON graph (Rules 1 and 5):

$$\text{Topo-QoS}(v) = 0.6 \cdot \text{BT}_w(v) + 0.4 \cdot \text{AP}(v), \quad (5)$$

where BT_w is betweenness over edge distances $d(e) = 1/(w(e) + 10^{-6})$, so tightly coupled edges attract shortest paths, and AP flags articulation points. Because of an implementation defect, the articulation term reads zero for every node, so the reported score ranks as QoS-weighted betweenness. Restoring the term lowers Topo-QoS from 0.553 to 0.533, so the defect does not favor the reference counts. We archive the unweighted Topo baseline in the replication repository.

Three primary training-free rankings restate the reachability oracle’s propagation rule and are reported as references: the first-order expansion of I^* (Analytic I^* , Eq. (6)); InDeg(v), the number of direct dependents of v on the Application–Library DEPENDS_ON graph, which for an Application equals publish–subscribe afferent coupling and is exactly the set I^* ’s first wave reaches (Remark 1); and Reach(v), the number of its transitive dependents normalized by $|V| - 1$, which is the set a reachability cascade can visit. In addition, we archive raw-multigraph topic counts (Pubs-raw) and transitive closure without Rule 5 (Reach-R1) in the replication repository. A reference shows how much of an oracle a restatement of its rule recovers, which is the size of the circularity, not predictive skill. References carry no contrast against Topo-QoS, and a learner’s distance from them is reported as such, not as a test.

5.3. Metrics, Protocols and Statistics

Population. Every predictor is scored on the Application set V_{app} .

Metrics. We use Spearman ρ to measure ranking against the simulation oracles. To distinguish between active components and those with zero impact, we report both the full-population Spearman ρ and the active-stratum Spearman $\rho_{>0}$, which only includes components with true impact greater than zero. To identify critical sets, we use $\text{Overlap}@K$, the fraction of the true top- K recovered by the predicted top- K , where K is about 20% of $|V_{\text{app}}|$. We also report PR-AUC, $F_1@ \tau$, and nDCG@10 in the replication repository.

Protocols. In the LOSO setup, we train learned models on eleven scenarios and test on the twelfth, repeating this for all 12 folds and five random seeds. The reported value is the average ρ across seeds. For zero-shot transfer, we evaluate models built on all twelve scenarios across the five system models without any fine-tuning. Training-free rankers are not fitted to any data, so for them, both protocols become per-scenario analyses on the same twelve scenarios and five models. We include them in the same tables to ensure every row is scored on the same scenarios and labels.

Statistics. The summary ρ is the average Spearman correlation for each fold or system, with a 95% confidence interval from a bootstrap that resamples folds ($B = 2,000$) [72]. We compare results using paired two-sided Wilcoxon signed-rank tests across folds [73]. With 12 folds, the smallest possible two-sided p -value is 0.00049. Fisher- z averaging does not change the row order in Table 5. Weighting folds by $|V_{\text{app}}|$ keeps the top five the same (the reference **InDeg** rises to 0.810) but increases **Topo-QoS** to 0.596, corresponding to the raw-multigraph learned engines (0.595–0.606). Spearman ρ gives tied scores their average rank. $\text{Overlap}@K$ breaks ties at the K boundary by node order, which is arbitrary for integer counts, so Section 6.1 also reports tie-aware recall curves. Because LOSO folds share ten of eleven training scenarios, we also test every registered contrast with the Nadeau–Bengio corrected resampled t -test [74] and, when weighting by $|V_{\text{app}}|$, with an exact sign-flip permutation test.

Analysis plan and the status of each result. The evaluation protocol is version-controlled in the replication repository. We registered fourteen numbered amendments across the study, encompassing sensitivity, ablation, and reclassification arms, with none withheld. We assign every reported result to one of three tiers. *Confirmatory*: the original plan’s primary contrast (**HGT-QoS** vs. **Topo-QoS**) and unweighted contrast (**HGT** vs. **Topo-QoS**), Holm-corrected across the two. *Registered secondary*: hypotheses committed before their respective arms executed—including the matched 2×2 , hybrid engines, reference counts, dependency-graph learners, derivation value, full-population queue-flow labels, and sensitivity arms, each Holm-corrected within its family. We conducted all secondary analyses after observing the primary null. *Exploratory*: post hoc investigations and reference reclassifications (which altered no figures; Section 4.4). **Plan deviation**: the registered nested inner selection was simplified to a fixed configuration during primary sweeps; a sensitivity analysis assesses the registered selection (Section 6.2). Because sequential amendment registration accumulates family-wise error throughout rounds, we also evaluate an omnibus Holm correction across all 13 decision-bearing contrasts (Supplementary Table S23; `reproduce/omnibus_holm.py`). Both hybrid contrasts against **Topo-QoS** remain significant under this omnibus adjustment (Hybrid-GAT $p_{\text{omni}} = 0.019$, Hybrid-HGT $p_{\text{omni}} = 0.041$). In contrast, the primary contrast remains null (+0.069, nominal $p = 0.266$, $p_{\text{omni}} \geq 0.46$). Main-text tables report per-family Holm corrections to reflect the pre-specified hypothesis families.

6. Results

All results are reported on the Application population (V_{app}) against the primary oracle $I^*(v)$, and, for the rankers of Table 6, also against the queue-flow oracle $I_{\text{dyn}}(v)$ and the multi-criteria oracle $I_{\text{comp}}(v)$ (Section 4.3). Only the primary and secondary contrasts are confirmatory; every other contrast is registered as secondary or exploratory (Section 5.3). Per-fold results, secondary strata, and extended protocol notes are in the public experiment pages (Section 5.1). Figure 4 summarizes the main findings.

6.1. RQ1: Ranking Accuracy of Learned Engines

Summary. Using derived dependency graphs allows for highly accurate cascade-impact ranking. The graph attention network (**GAT-P-QoS**) achieves $\rho = 0.748$ against the primary reachability oracle I^* . Causal ablation tests show this strong performance comes from representing the degree distribution (losing 0.14 without degree

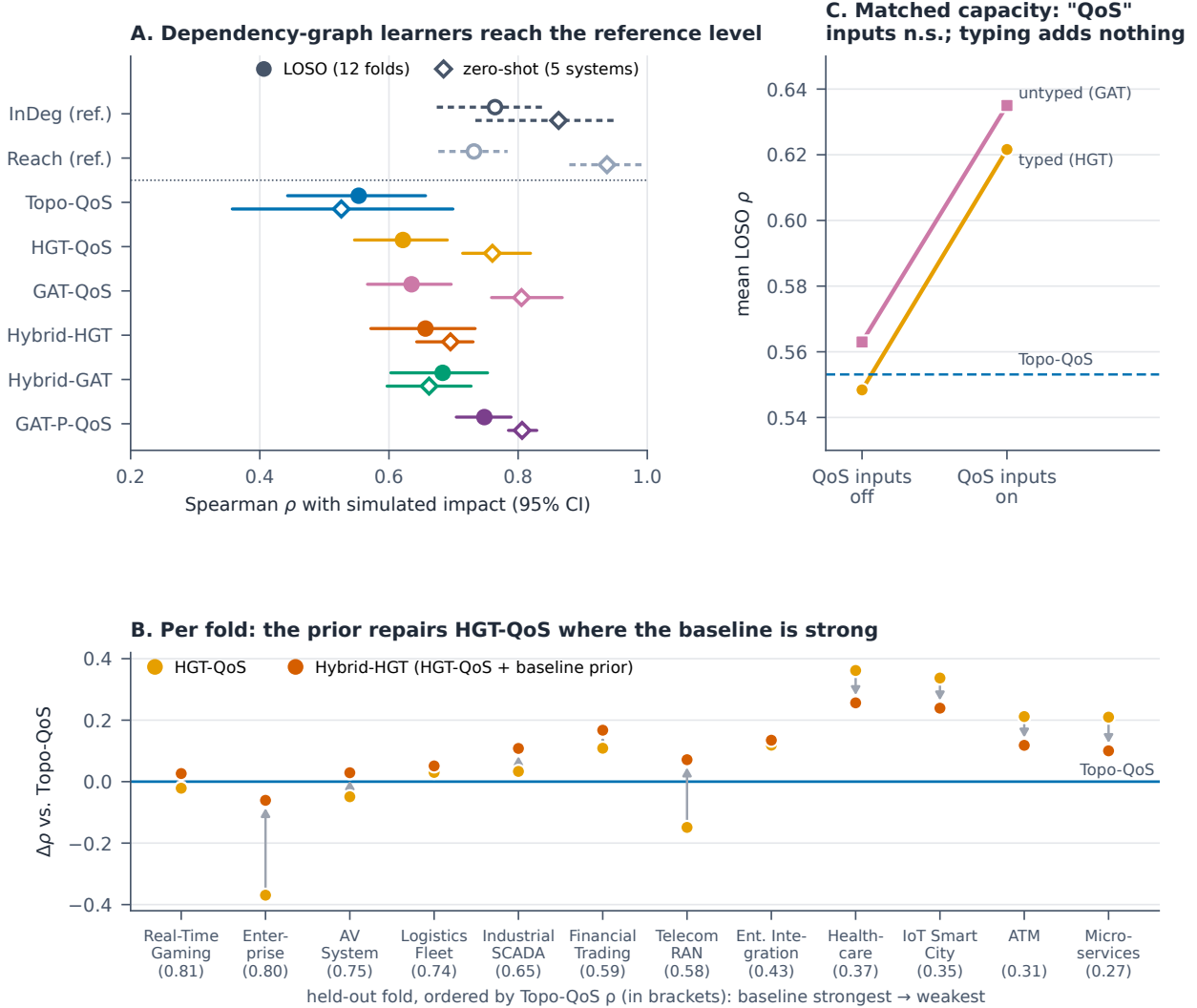


Figure 4: Main results for the Application population. (A) Shows the mean Spearman ρ with 95% bootstrap intervals under LOSO (filled circles; see Table 5) and zero-shot results on five system models (open diamonds). Grey hollow markers represent reference rankings that restate I^* 's propagation rule (InDeg, Reach) and are not predictors. The neural model using the dependency graph (GAT-P-QoS) performs similarly to InDeg; the two are statistically similar but not identical. (B) Shows the improvement of HGT-QoS and Hybrid-HGT over Topo-QoS for each held-out fold. (C) Displays cell means from the capacity- and channel-matched 2×2 (Table 7): the "QoS" inputs, which include a QoS-weighted in-degree column, increase both models by about 0.07 (not significant after Holm correction), while typed and untyped configurations show no difference. When w_{in} is held in both arms, the "QoS" effect drops to +0.030 (Table 8).

features; see Section 6.2), corresponding to the first-order reference level (0.764, statistically similar but not equivalent within ± 0.05). For the more expensive queue-flow oracle I_{dyn} (12.7 CPU-hours), a learned surrogate using dependency-graph features reaches $\rho = 0.799$, considerably outperforming all closed-form approximations (0.706, Holm $p = 0.019$) and capturing declared QoS signals (+0.095). Under multi-criteria failure conditions (I_{comp}), hybrid engines remain stable (0.58) while pure neural models drop sharply (0.144–0.334). On the raw multigraph, the main registered comparison (HGT-QoS vs. **Topo-QoS**, 0.553; corrected 0.533) showed no significant difference ($\Delta\rho = +0.069$, $p = 0.266$). To recover 80% of the true top-20% set, **GAT-P-QoS** must flag the top 40%.

Learned engines are trained on eleven scenarios and scored on the twelfth; training-free rankers are scored on all twelve (26 to 300 Applications, K between 5 and 60). Table 5 reports all of them against I^* , with the reference rankings of Section 5.2 in a separate block: they restate I^* ’s rule, so their rows measure the circularity and carry no contrast.

Table 5: Main results under LOSO across twelve synthetic architectures (Application population; learned engines: five seeds, CPU sweeps). $\Delta\rho$ is paired by fold against **Topo-QoS**, with a bootstrap 95% CI ($B = 2,000$) and a two-sided Wilcoxon signed-rank test. $\rho_{>0}$ denotes Spearman correlation restricted to active components ($I^* > 0$). p_{Holm} is given within each registered family: hybrids (registered secondary). The reference block holds canonical rankings that restate I^* ’s propagation rule (the post hoc first-order expansion, Eq. (6); the dependency counts); they are not predictors and carry no contrast. [‡]Exploratory: contrasts against the references and raw counterparts. Note: **Topo-QoS** carries an implementation defect (articulation term zero; Section 5.2); corrected value is 0.533. Additional training-free baselines and exploratory raw-graph references (Pubs-raw, Reach-R1, Degree-raw, RevPR-raw, Topo), per-fold values, contrasts among references, and **GAT-P+InDeg** are provided in the replication repository. Underlined: best predictor per column.

Predictor	LOSO ρ [95% CI]	Active $\rho_{>0}$	$\Delta\rho$ vs Topo-QoS [95% CI]	Won	p (p_{Holm})	Overlap@ K
<i>Reference: restatements of I^*’s propagation rule (not predictors)</i>						
Analytic I^*	0.808 [0.755, 0.853]	0.631	—	—	—	0.536
InDeg (direct dependents)	0.764 [0.674, 0.840]	0.516	—	—	—	0.504
Reach (transitive dependents)	0.732 [0.674, 0.782]	0.286	—	—	—	0.344
<i>Training-free baseline (registered comparator)</i>						
Topo-QoS	0.553 [0.443, 0.657]	0.280	—	—	—	0.388
<i>Learned, on the dependency graph</i>						
GAT-P-QoS	0.748 [0.704, 0.789]	0.440	+0.195 [+0.100, +0.294]	10/12	0.0068 [‡]	0.454
HGT-P-QoS	0.514 [0.392, 0.636]	0.237	−0.039 [−0.155, +0.077]	4/12	0.622 [‡]	0.380
<i>Learned and hybrid, on the raw multigraph</i>						
HGT-QoS	0.622 [0.547, 0.690]	0.312	+0.069 [−0.046, +0.174]	8/12	0.266	0.426
GAT-QoS	0.635 [0.567, 0.696]	0.338	+0.082 [−0.046, +0.201]	7/12	0.233	0.438
Hybrid-HGT	0.657 [0.572, 0.733]	0.345	+0.103 [+0.055, +0.152]	11/12	0.0034 (0.0068)	0.435
Hybrid-GAT	0.683 [0.603, 0.753]	0.362	+0.130 [+0.075, +0.190]	11/12	0.0015 (0.0029)	0.450

The reference level. As noted in Remark 1, **InDeg** represents publish–subscribe afferent coupling [44, 45, 46] and counts exactly the Applications reached by the first propagation wave of I^* . **Reach** counts the transitive dependents that a reachability cascade can visit. Their values on I^* (0.764, $\rho_{>0} = 0.516$; 0.732) show how much of the oracle is recovered by expressing again its rule, rather than measuring predictive skill. Exploratory raw-multigraph references are evaluated in the replication repository: counting published topics (Pubs-raw) achieves 0.731, and transitive closure without Rule 5 (Reach-R1) gets 0.674. Among the references, that derivation matters in one case: without Rule 5, **Reach** drops by 0.058 (9/12 folds, Holm $p = 0.0068$; derivation ablation). Untyped raw-graph metrics do not recover the reference: total degree reaches 0.199 and reverse PageRank 0.089, while PageRank is constant for every Application because they have no incoming raw edges.

The first-order expansion of the oracle. The strongest reference is a post hoc closed-form first-order expansion of I^* :

$$\hat{I}_1^*(v) = \sum_{t \in \text{pub}(v)} \frac{|\text{sub}(t)|}{|\text{pub}(t)|}, \quad (6)$$

where $|\text{pub}(t)| \geq 1$ holds for all published topics $t \in \text{pub}(v)$ by construction (v publishes to t), preventing any division by zero. It reaches mean $\rho = 0.808$ [0.755, 0.853] ($\rho_{>0} = 0.631$): direct subscriber loss is most of what I^* measures, and **InDeg** is a coarser version of the same quantity. As a reference, it bounds the mean

rather than every fold: on four folds (AV, Financial Trading, Healthcare, Industrial SCADA), **InDeg** slightly exceeds it (by 0.005–0.038).

Learners on the dependency graph approach the reference. The same graph neural networks, after being trained on the dependency graph instead of the raw multigraph, achieve $\rho = 0.748$ (**GAT-P-QoS**), making it the best GNN in Table 5. This is 0.016 below the direct-dependent count for individual seeds, and the five-seed ensemble is 0.007 above it (0.772, Table 6). These differences are not statistically significant, nor do they show equivalence: the 90% interval of the per-fold difference is ± 0.076 , so a two one-sided test at the registered margin of ± 0.05 does not pass ($p = 0.17$ per seed, 0.13 for the ensemble; equivalence arm F3). At this sample size, the learner and the count are indistinguishable and not ranked. The closeness to the reference depends on the degree features each learner receives (Section 3.5): removing **in_degree** and w_{in} lowers **GAT-P-QoS** to 0.612 (Section 6.2). We archive a version that uses **InDeg** as its prior in the replication repository, since it is a reference prior. **HGT-P-QoS** only reaches 0.514 and is highly dependent on the seed (mean within-fold seed SD 0.254; Section 7.3).

Hybrid engines secure statistically significant gains beyond baseline priors. Hybrid-HGT achieves $\rho = 0.657$ ($+0.103$, Holm $p = 0.0068$) and Hybrid-GAT reaches 0.683 ($+0.130$, Holm $p = 0.0029$) compared to **Topo-QoS**, each outperforming the baseline in 11 out of 12 folds. Both results remain statistically significant under the omnibus Holm correction ($p_{\text{omni}} = 0.041$ and 0.019), the Nadeau–Bengio corrected t ($p = 0.019$ and 0.014), and $|V_{\text{app}}|$ -weighted averaging (exact sign-flip $p = 0.038$ and 0.005). In this comparison, a hybrid uses $p(v)$ as an input and adds a learned correction to its logit, nesting its comparator. When compared to their own base learners, neither hybrid shows a significant difference (Hybrid-HGT vs. **HGT-QoS** $+0.035$, $p = 0.73$; Hybrid-GAT vs. **GAT-QoS** $+0.048$, $p = 0.30$). The prior also reduces transfer performance: in zero-shot tests, both hybrids perform worse than their base learners (Section 6.3). On their own, the raw-multigraph learned engines do not differ significantly from **Topo-QoS** (**HGT-QoS** $+0.069$, $p = 0.266$; **GAT-QoS** $+0.082$, $p = 0.233$).

Table 6: Rankers against three oracles, twelve LOSO folds, Application population (registered secondary and exploratory). All three oracles are exhaustive over the 1,321 Applications; I_{dyn} is the five-seed mean across all components. Partial ρ : Spearman correlation with I_{dyn} after the rank of I^* , or of the first-order expansion $\hat{I}_1^*$ (Eq. (6)), is regressed out of both. Published learned rows score the mean of the five seeds’ predictions, which raises ρ on I^* above the per-seed means of Table 5; the GBM surrogate rows and GAT surrogate are per-seed means. *Trained on I_{dyn} labels of the eleven training scenarios: a surrogate for I_{dyn} , not an independent check of it. Reference rows restate I^* ’s rule (Section 4.4). (For **Topo-QoS**, the corrected I^* value is 0.533; Section 5.2. Rate-weighted analytical expansion $\hat{I}_{\text{dyn},1}^{\text{rate}}$ yields $\rho = 0.830$ without training; Section 7.2.) Underlined: highest value per column among predictors. 95% CIs are bootstrapped across folds.

Ranker	$I^* \rho$	$I_{\text{dyn}} \rho$ [95% CI]	Partial $\rho(\cdot, I_{\text{dyn}} I^*)$ [95% CI]	Partial $\rho(\cdot, I_{\text{dyn}} \hat{I}_1^*)$	$I_{\text{comp}} \rho$
<i>Reference: restatements of I^*’s propagation rule (not predictors)</i>					
Analytic I^*	0.808	0.706 [0.629, 0.783]	0.318 [0.234, 0.412]	—	0.636
InDeg	0.764	0.664 [0.562, 0.758]	0.272 [0.188, 0.366]	0.079	0.650
Reach	0.732	0.583 [0.510, 0.651]	0.117 [0.055, 0.184]	0.102	0.302
<i>Training-free baseline</i>					
Topo-QoS	0.553	0.471 [0.381, 0.559]	0.134 [0.061, 0.204]	−0.091	<u>0.702</u>
<i>Learned GNNs (seed ensemble)</i>					
GAT-P-QoS	0.772	0.615 [0.549, 0.673]	0.111 [0.016, 0.213]	0.182	0.274
Hybrid-GAT	0.702	0.599 [0.514, 0.669]	<u>0.173</u> [0.082, 0.267]	0.056	0.585
Hybrid-HGT	0.672	0.573 [0.497, 0.638]	0.162 [0.074, 0.252]	0.029	0.582
HGT-QoS	0.667	0.549 [0.464, 0.622]	0.130 [0.035, 0.224]	0.212	0.201
GAT-QoS	0.647	0.523 [0.436, 0.606]	0.107 [0.018, 0.199]	<u>0.217</u>	0.144
HGT-P-QoS	0.618	0.496 [0.395, 0.601]	0.112 [0.033, 0.190]	0.087	0.334
<i>Dynamic queue-flow surrogates (trained on I_{dyn})</i>					
GBM-P-QoS→dyn*	<u>0.764</u>	<u>0.799</u>	—	—	0.477
GAT-P-QoS→dyn*	0.734	0.598	—	—	0.368

Beyond the reachability oracle. Table 6 compares the rankers against two other oracles. On the full-population I_{dyn} , which matches I^* at $\rho = 0.711$ [0.625, 0.794], the reference methods outperform all GNNs (first-order expansion 0.706, **InDeg** 0.664, compared to 0.496–0.615 for GNNs), and **Topo-QoS** achieves 0.471. Since I_{dyn} shares its topology and first-order effect with I^* , the partial correlation shows how much of

each ranking remains after removing the oracle it was fitted to or restates. After removing the rank of I^* , InDeg keeps 0.272 [0.188, 0.366], Reach 0.117, Topo-QoS 0.134, and the GNNs 0.107–0.173, with all intervals excluding zero. If we remove the first-order expansion instead, the order reverses: InDeg stays at 0.079. Meanwhile, the raw-multigraph GNNs keep most of their value (0.212–0.217), showing that the learners capture queue-flow signals beyond what the direct count provides. Learning from the right labels is more important than the graph model itself. A gradient-boosted model using counts, the first-order expansion, and QoS features, trained on I_{dyn} labels, ranks held-out architectures at 0.799, outperforming the first-order expansion (+0.094, Holm $p = 0.019$) and InDeg (+0.135, Holm $p = 0.019$; Family B). Its QoS features provide this gain (+0.095 over the same model without them, Holm $p = 0.014$), while on I^* they add nothing (−0.003, $p = 0.62$; Family C). The GNN trained on the same labels does not: GAT-P-QoS→dyn reaches 0.598, below the first-order expansion (−0.108, Holm $p = 0.014$) and the gradient-boosted surrogate (−0.201, Holm $p = 0.0015$), and is no better than the same GNN trained on I^* (−0.008; control F5). Importantly, this comparison has a known feature difference: the tree surrogate had access to the analytical first-order expansion $\hat{I}_1^*$, declared publication rates, and message payload sizes, while the GAT surrogate only used topological graph features and did not have access to rates, payloads, or the analytical baseline. The tree model’s +0.094 advantage over closed-form approximations comes from using rate and payload data, not just from tree inductive biases over neural graph architectures (Section 7.2); when publication rates are directly incorporated into an analytical expansion ($\hat{I}_{\text{dyn},1}^{\text{rate}}$), it achieves a mean $\rho = 0.830$ across the twelve scenarios without training, showing that rate-weighted closed forms match or exceed the surrogate when rate physics can be directly expressed. These surrogates are trained on the oracle they are evaluated on; their main value is cost, since labeling I_{dyn} took 12.7 CPU-hours for the corpus (Section 6.4). Upon training on I^* instead, the same gradient-boosted model is less robust across the three oracles than the first-order expansion alone (worst-case $\rho -0.121$, Holm $p = 0.001$; Family A).

The multi-criteria oracle favors different rankers. On I_{comp} , which includes fragmentation, throughput, and flow disruption (Section 4.3), learned engines without a prior perform poorly (0.144–0.334; sum-aggregation GNNs from Section 6.2 reach 0.41–0.44). The two hybrids that use Topo-QoS as a prior maintain 0.58. Training-free scores that correspond to the oracle’s terms rank higher: Topo-QoS achieves 0.702 [0.649, 0.753] (higher than the direct-dependent reference at 0.650; the reference is ahead on only 3 of 12 folds, $p = 0.110$), and raw total degree reaches 0.719. Reach drops to 0.302. This lead reflects overlap with I_{comp} ’s fragmentation and flow terms, similar to the references’ lead on I^* (Section 4.4), rather than predictive skill. No ranker is best across all three oracles.

Identifying the critical set. Rank correlation across all Applications is not the main concern in a CI gate; the key question is whether the true top-ranked items are flagged. Figure 5 shows, for each ranker, the proportion of the true top-20% set on I^* that appears within its top $k\%$, with ties resolved by averaging. At $k = 20\%$, GAT-P-QoS recovers 0.48 of the critical set and Topo-QoS recovers 0.38, matching their Overlap@ K (0.454 and 0.388). This means about half of what a top-20% gate flags is not in the true top 20%, and about half of the true top 20% is missed. To recover 80% of the critical set, GAT-P-QoS must flag the top 40% (0.81), and Topo-QoS does not reach 80% within 50%. The reference methods do only slightly better: at $k = 20\%$, InDeg recovers 0.49 (tie-breaking bounds 0.48–0.50), and 80% recall requires the top 45% by InDeg (0.83) or by the first-order expansion (0.89). Only the first-order expansion reaches 90% within the top 50%, and Reach does not reach 80% within 50%. In the earlier developmental sample ($I_{\text{dyn}}^{n=30}$; right panel of Figure 5), GAT-P-QoS reaches 0.65 at 40% and 0.80 at 50%, compared to 0.83 at 40% for the InDeg reference. In practice, requiring a gate to flag 40% to 50% of components to catch 80% of critical points would cause significant developer alert fatigue. This suggests that SaG is best used as a prioritized triage list for architectural review, not as an automated blocking gate (the two-tiered triage protocol to reduce alert fatigue is described in Section 7.1).

6.2. RQ2: What Learned Engines Need

Summary. Learned engines improve by +0.08 to +0.11 when using the dependency graph, where messages get the scored Applications from their dependents (in 11–12 out of 12 folds). The dependency-graph GAT’s closeness to the reference comes from its degree features: without them, it loses 0.14 (Holm $p = 0.005$). A sum-aggregation GNN, which can count dependents, loses only 0.01 without these features, though the

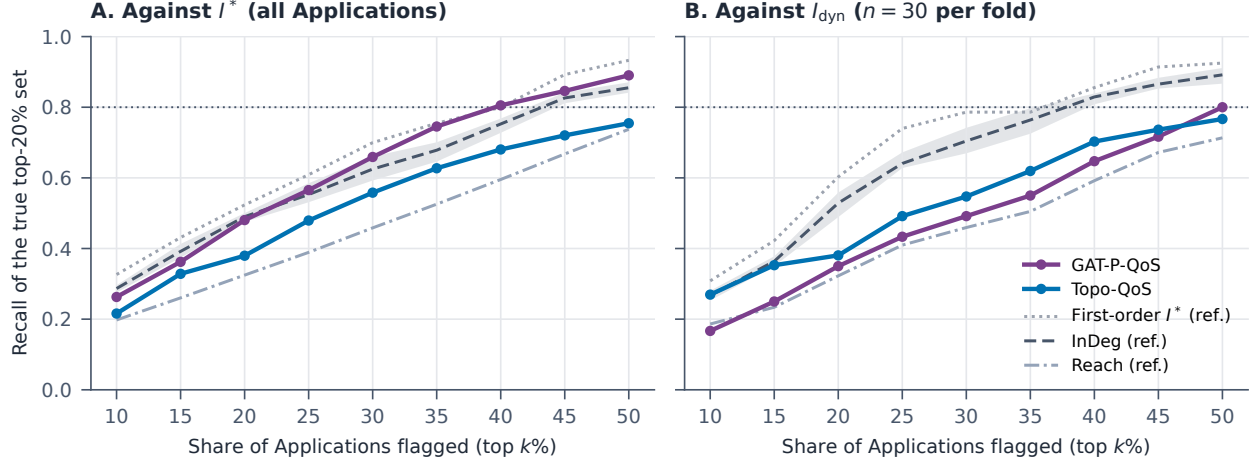


Figure 5: Recall of the true top-20% set (tie-inclusive) by each ranker’s top $k\%$, mean over the twelve LOSO folds, against I^* (left) and the exploratory developmental sample ($I_{\text{dyn}}^{n=30}$; right; Table 6 reports the exhaustive $N = 1,321$ population). Ties are resolved in expectation over random tie-breaking; grey dashed curves are reference rankings that restate I^* ’s rule, and the shaded band gives InDeg’s optimistic and pessimistic bounds. Reach, whose scores tie heavily, has much wider bounds (0.12–0.80 at $k = 20\%$ on I^*), so its expected curve is the only meaningful summary. The dotted line marks 80% recall.

614 difference is not significant after correction. When capacity is matched, relation typing has no effect; with
 615 weighted in-degree included in both arms, the “QoS” inputs also add no measurable benefit (+0.030, Holm
 616 $p = 0.33$).

Table 7: The 2×2 factorial design with capacity and edge-channel width matched: GAT ($\neg T \neg Q$, 437,496 parameters), HGT ($T \neg Q$, 434,620), GAT-QoS ($\neg T Q$, 429,992), HGT-QoS ($T Q$, 434,620). One CPU sweep, twelve LOSO folds, five seeds, Application population. Holm correction across the three orthogonal quantities. Cell means: 0.563, 0.548, 0.635, 0.622.

Quantity	Contrast	$\Delta\rho$	95% CI	Won	W	p	p_{Holm}
<i>Three orthogonal quantities, Holm-corrected across these three</i>							
Typing (main effect)	averaged over Q	−0.014	[−0.052, +0.023]	4/12	29.0	0.470	0.940
“QoS” inputs (main effect)	averaged over T	+0.073	[+0.013, +0.120]	10/12	13.0	0.043	0.127
Typing $\times$ “QoS” interaction	difference of differences	+0.001	[−0.050, +0.042]	6/12	34.0	0.733	0.940
<i>Simple effects — descriptive, not separately corrected</i>							
Typing, QoS absent	HGT vs. GAT	−0.015	[−0.064, +0.033]	5/12	31.0	0.569	—
Typing, QoS present	HGT-QoS vs. GAT-QoS	−0.013	[−0.054, +0.026]	4/12	27.0	0.380	—
“QoS” inputs, typing absent	GAT-QoS vs. GAT	+0.072	[+0.028, +0.109]	10/12	9.0	0.016	—
“QoS” inputs, typing present	HGT-QoS vs. HGT	+0.073	[−0.002, +0.136]	10/12	19.0	0.129	—

617 **The “QoS” factor was mostly the weighted in-degree.** In Table 7, the main effect of QoS (+0.073)
 618 is not significant after Holm correction ($p_{\text{Holm}} = 0.127$), and it cannot be attributed to QoS content. w_{in} ,
 619 the sum of incoming dependency weights, is a QoS-weighted version of InDeg, and the QoS-off arms set it
 620 to zero. A sensitivity analysis repeated the matched 2×2 with w_{in} included in the QoS-off cells (Table 8,
 621 F4): the main effect drops to +0.030 [−0.023, +0.085] (Holm $p = 0.33$), typing is −0.036, and the interaction
 622 is +0.046, none of which are significant. This means more than half of the earlier QoS effect was due to
 623 the weighted in-degree. The closed-form controls show the same pattern: unweighted betweenness (0.591)
 624 and constant topic weights (0.595) match or exceed QoS-weighted betweenness (0.553), and permuting QoS
 625 profiles across topics has no effect ($\Delta = -0.006$ [−0.025, +0.015], $p = 0.733$). On I^* , there is no evidence that
 626 QoS contract content affects ranking, but on I_{dyn} , which uses the contracts, it does (Section 6.1, Family C).
 627 **What the dependency-graph learner needs.** Each learner receives in-degree and w_{in} as node properties,
 628 and on the dependency graph, it also aggregates the dependent neighborhood. This means its similarity
 629 to InDeg could come from either source. Ablation experiments (Table 8) separate these effects. Removing
 630 the two degree columns reduces GAT-P-QoS by 0.136, and removing the three centralities most correlated
 631 with in-degree reduces it by 0.161 (F1, rule F1a). This shows that the approach to the reference depends
 632 on the degree features. Message-passing theory predicts this: softmax attention normalizes over neighbors

Table 8: Sensitivity and ablation arms: twelve LOSO folds, five seeds, one CPU invocation with every comparator re-run (each reproduces its published per-seed ρ exactly), Application population. $\Delta\rho$ is paired by fold against the comparator named, with a bootstrap 95% CI; Holm within each registered family (F1: degree features; F2: aggregator; F4: the 2×2 with w_{in} held). $-\text{deg}$: `in_degree` and w_{in} zeroed; $-\text{deg}^*$: also PageRank, closeness and eigenvector centrality, whose correlation with `InDeg` is 0.74–0.77 on average. Zero-shot: mean ρ on the five system models. Registered secondary.

Arm	Comparator	LOSO ρ (I^*)	$\Delta\rho$ [95% CI]	Won	p_{Holm}	$I_{\text{dyn}} / I_{\text{comp}}$	Zero-shot
<i>F1: removing the degree features</i>							
GAT-P-QoS–deg	GAT-P-QoS (0.748)	0.612	−0.136 [−0.205, −0.072]	1/12	0.0049	0.485 / 0.156	0.821
GAT-P-QoS–deg*	GAT-P-QoS	0.587	−0.161 [−0.252, −0.080]	1/12	0.0049	0.462 / 0.138	0.654
GAT-QoS–deg	GAT-QoS (0.635)	0.365	−0.270 [−0.344, −0.199]	0/12	0.0015	0.295 / −0.047	0.810
<i>F2: sum aggregation (Graph Isomorphism Network with Edge features — GINE, 434,123 parameters) instead of softmax attention</i>							
GIN-P-QoS	GAT-P-QoS	0.732	−0.016 [−0.055, +0.017]	7/12	0.910	0.606 / 0.407	0.806
GIN-P-QoS–deg	GAT-P-QoS–deg	0.721	+0.108 [+0.027, +0.197]	8/12	0.157	0.602 / 0.428	0.810
GIN-P-QoS–deg*	GAT-P-QoS–deg*	0.711	+0.124 [+0.031, +0.230]	9/12	0.157	0.589 / 0.438	0.776
<i>F4: matched 2×2 with w_{in} held (cells GAT+w_{in} 0.628, HGT+w_{in} 0.569, GAT-QoS 0.635, HGT-QoS 0.622)</i>							
Typing (main effect)	averaged over Q	—	−0.036 [−0.079, +0.004]	2/12	0.330	—	—
“QoS” inputs (main effect)	averaged over T	—	+0.030 [−0.023, +0.085]	9/12	0.330	—	—
Interaction	difference of differences	—	+0.046 [−0.009, +0.101]	9/12	0.330	—	—

and computes a weighted mean, so it cannot count them, whereas a sum aggregator can [59, 60]. A GINE network with the same parameter budget supports this, keeping 0.721 and 0.711 without the same columns, only 0.01–0.02 below its full-feature score. However, the aggregator contrasts are not significant after Holm correction (+0.108 and +0.124, $p_{\text{Holm}} = 0.157$; rule F2b), so the study does not claim an aggregator effect. No dependency-graph learner, with or without degree features, is equivalent to `InDeg` within ± 0.05 (equivalence bounds 0.063–0.276; F3). In zero-shot, the degree columns matter little (0.810–0.821 without them), because the system models most of the ranking by separating inert from active components (Section 6.3).

Relation typing under matched capacity. HGT-QoS is within 0.013 of GAT-QoS (Table 7). On the raw multigraph, this does not say much about typing, since no message gets an Application there: a gradient-boosted model using the same per-node features but no graph (**GBM-Feat**) achieves 0.642, similar to GAT-QoS, and removing HGT’s reverse pass (the only way messages get Applications) has no effect (HGT-QoS-U, −0.010, $p = 0.91$). On the dependency graph, where messages do reach Applications, HGT-P-QoS gets 0.514 with a mean within-fold seed SD of 0.254, compared to 0.748 and 0.030 for GAT-P-QoS. This is for one fixed configuration (width 100), and averaging predictions from five seeds raises it to 0.618. We interpret this as seed instability for that configuration, not as evidence against relational typing in general. **The registered selection rule.** The plan selected every hyperparameter by nested selection, which was not used in the published results (Section 5.3). When applied later on the stage-1 grid (arm N), it changes HGT-QoS from 0.622 to 0.677 (+0.055) and GAT-P-QoS from 0.748 to 0.693 (−0.054), neither of which is significant (Holm $p = 0.259$). Nested HGT-QoS versus **Topo-QoS** is +0.123 on 9 of 12 folds, still not significant (Holm $p = 0.192$; rule F6b). Using the registered rule does not change the main result. Table 5 shows the fixed, published configuration; Arm N is a registered secondary sensitivity analysis. The search harness also stops early on a held-out training scenario instead of the published node-level split. On the three folds where it chose the published hyperparameters, it does not reproduce the published scores (differences up to 0.10), so these differences are not due to hyperparameter selection alone.

6.3. RQ3: Zero-Shot Transfer to Models of Open-Source Systems

Summary. On five hand-crafted models of open-source systems, the best learned engines achieve about 0.81 zero-shot ranking for I^* (GAT-P-QoS 0.806, GAT-QoS 0.805), much higher than the training-free baseline (0.526). However, their agreement on the active stratum is weak ($\rho_{>0} \leq 0.342$). The reference rankings are even higher (transitive dependents 0.938, $\rho_{>0} = 0.871$; direct dependents 0.863), since in these models, most of I^* is just reachability. These models are structurally dissimilar; the author wrote all five, so these results reflect transfer to these models, not to real-world systems. The baseline prior reduces transfer: both hybrids perform worse than their base learners (0.695 and 0.662 compared to 0.760 and 0.805).

The **Reach** reference achieves $\rho = 0.938$ and $\rho_{>0} = 0.871$ on the five models: 0.836–0.997 on the three original publish–subscribe systems and 0.966–0.998 on the two RPC systems converted to pub-sub meshes. These values are higher than anything **Reach** achieves on the synthetic folds (0.732, $\rho_{>0} = 0.286$), and this

Table 9: Zero-shot transfer to hand-authored models inspired by five open-source systems (Application population). Models are evaluated out-of-distribution without fine-tuning. $\rho_{>0}$ denotes rank correlation restricted to active components ($I^* > 0$). PR-AUC measures critical-set identification quality. All rows are scored on the same labels (the learned engines’ zero-shot label files; zero-shot rule R5). The training-free rows are not trained on anything, so “zero-shot” applies only to the learned rows. Reference rows restate I^* ’s propagation rule and are not predictors. Underlined: best predictor per column. 95% CIs are bootstrapped across the five systems and are indicative only.

Predictor	Evaluation Substrate	Mean ρ [95% CI]	Active $\rho_{>0}$	PR-AUC
<i>Reference: restatements of I^*’s propagation rule (not predictors)</i>				
Reach	Dependency graph	0.938 [0.879, 0.991]	0.871	0.933
InDeg	Dependency graph	0.863 [0.734, 0.952]	0.321	0.752
<i>Training-free baseline</i>				
Topo-QoS	Dependency graph	0.526 [0.357, 0.699]	−0.088	0.474
<i>Learned and hybrid, raw multigraph</i>				
HGT-QoS	Raw multigraph	0.760 [0.714, 0.819]	0.236	0.713
GAT-QoS	Raw multigraph	0.805 [0.759, 0.868]	0.319	0.790
Hybrid-HGT	Raw multigraph	0.695 [0.643, 0.730]	0.210	0.602
Hybrid-GAT	Raw multigraph	0.662 [0.597, 0.727]	0.185	0.600
<i>Learned, dependency graph</i>				
GAT-P-QoS	Dependency graph	<u>0.806</u> [0.785, 0.829]	<u>0.342</u>	<u>0.838</u>

difference is due to structural factors. Half of the system models’ Applications have zero simulated impact (0.51 on average, compared to 0.31 on the folds), and their fan-in is more concentrated (Gini 0.65 vs. 0.50). On these graphs, simply separating the inert half from the rest already gives a high full-population ρ , and **Reach**, which is zero for components with no dependents, does this perfectly. The active-stratum $\rho_{>0}$ is more informative here. On this measure, all predictors are weak (learned engines 0.185–0.342, training-free baseline negative), as is the direct-dependent reference (0.321 [−0.044, 0.687]), while **Reach** (0.871) is much higher. In these models, ordering active components by transitive reach closely matches what I^* computes, showing strong circularity rather than transferable skill. Because all five models were created by a single author without independent cross-validation, and two are stylized pub-sub versions of synchronous RPC systems (Section 5.1), these results show that learned rankers can transfer broad topological partitioning across hand-written model styles, but do not provide reliable within-cascade discrimination on new architectures.

6.4. RQ4: Analysis Cost and Sustainability

Summary. Measured like for like, on the same graphs in one session (Table 10), one labeling pass of the reachability oracle over the Applications takes 0.01–0.72 s per corpus architecture, the counting path that restates its first wave 17–176× less (median 45×), and the feature extraction every learned engine needs 4.5–72.5× more (median 16.9×). On generated graphs of up to 5,000 components, the ordering holds, with features at 5.7–18.7× one pass. The queue-flow oracle is the expensive one: 12.7 CPU-hours to label the corpus. Bounding energy consumption via CPU package power (28 W TDP) shows that lightweight surrogates save up to ~355 Wh compared to dynamic simulation. In contrast, for reachability, static counting remains the greenest choice.

Table 10 replaces two comparisons that earlier revisions got wrong. They set the full detection gate (system-layer analysis plus 18 anti-pattern detectors) against the five-seed labeling sweep over three node types. They called that sweep “one run” of I^* (median 5.6×), and they read earlier profiling sessions against different graphs. When timed together, each stage keeps its place in the ordering. The counting path is the cheapest, one I^* pass over the scored population is next, and the app-layer feature extraction that the learned engines read is the most expensive, at 4.5–72.5× one pass on the corpus. The articulation and Connectivity Degradation Index phase accounts for 88–91% of it (Enterprise, and the 2,000-component graph), and it is largest on Enterprise, whose Rule 1 projection is nearly complete. Where the I^* ranking itself is wanted,

Table 10: Like-for-like cost, every region timed on the same graph in one session (CPU, one thread, median of 3 runs, once at 4,995 components, where the three-type sweep was not run). Count: Application–Library projection plus InDeg. One I^* pass: seed 42, Applications, the population every ranker is scored on. Sweep: the published labeling run, five seeds over Application, Broker, and Library nodes. Features: the app-layer analysis that produces the learned engines’ node features. Gate: the system-layer analysis, quality scoring, and 18 anti-pattern detectors, the stage behind the earlier ratio of the detection gate. Corpus row: minimum–median–maximum over the twelve folds.

$ V $	Count (ms)	One I^* pass (s)	Sweep (s)	Features (s)	Gate (s)	Pass : count	Features : pass
249	2.5	0.26	1.63	1.46	2.05	105×	5.7×
499	6.6	0.79	6.59	6.49	10.01	119×	8.2×
999	13.1	3.90	32.42	30.57	50.92	297×	7.8×
1,998	35.5	18.45	157.86	160.73	275.80	521×	8.7×
4,995	226.2	94.58	—	1,768.81	3,072.60	418×	18.7×
Corpus	0.4–15.6	0.01–0.72	0.09–4.98	0.07–52.25	0.16–81.48	17–45–176×	4.5–16.9–72.5×

running the oracle is therefore cheaper than approximating it with a learned engine at every size measured, and the count is cheaper still, by one to two orders of magnitude, because it restates the oracle’s first wave; its practical advantage is cost, not correctness over the oracle it approximates. In the replication repository, timing profiles show that inference itself is negligible (tens of milliseconds at 2,000 components), and that the count scales to 10,000 components (1.1 s) and **Reach** less well (23 s), because it computes a transitive closure; complete timing artifacts and execution scripts are in `docs/research/jss/experiments/rq4-cost.md`. The expensive oracle is I_{dyn} : labeling the corpus took 12.7 CPU-hours (about 7 s per fault run and seed), which is the case for the learned surrogate of Section 6.1, whose inference runs in seconds. Although direct hardware power counters (e.g., Running Average Power Limit — RAPL, or external physical power meters) were not instrumented, we can establish a theoretical upper-bound proxy on computational energy expenditure using the host CPU’s base package power (28.0 W Thermal Design Power — TDP for the 13th Gen Intel Core i7-1370P; `reproduce/energy_estimate.py`). Under this upper-bound model, generating labels for the 12.7 CPU-hour I_{dyn} simulation expended up to ~ 355.6 Wh (≈ 1.28 MJ). In contrast, executing static feature extraction across the entire corpus required only 106.9 s (an upper bound of 0.83 Wh / 2.99 kJ), and computing the training-free counting baseline took just 11.1 s total (0.086 Wh / 310 J). From a Green Software Engineering and Green CI/CD perspective [21, 22, 23], deploying heavy neural architectures or full dynamic simulations in continuous integration represents a high-overhead, energy-intensive path. Learning pays its computational cost, along with its carbon footprint, only when it displaces repeated executions of expensive simulation oracles like I_{dyn} , providing a 19× speedup and massive energy savings; for cheap reachability analysis, simple static graph queries remain the strictly greener choice.

7. Discussion, Threats to Validity, and Limitations

7.1. Practical Consequences

A Dual-Regime Architectural Strategy for Continuous Resilience. The usefulness of a ranker depends on how much it costs to approximate the oracle. If the oracle is inexpensive, like the reachability oracle I^* (0.01–0.72 seconds per pass over a corpus architecture’s Applications; see Section 6.4), practitioners can compute its ranking directly from the same manifest, without needing an operational profile. The main finding here is that, for reachability impact, publish–subscribe afferent coupling—a direct-dependent count used in architecture research for two decades [44, 45]—performs as well as any graph neural network tested, but at a much lower cost. This is because it repeats the oracle’s first wave (Section 4.4), so it is not credited with predictive skill; its practical use is as a low-cost substitute for I^* in inner loops and large architectures. If the oracle is expensive, like the queue-flow simulator I_{dyn} (12.7 CPU-hours to label the corpus), a learned surrogate is justified: a gradient-boosted model using dependency counts, first-order expansion, and QoS features, trained on I_{dyn} labels from other architectures, ranks held-out architectures at $\rho = 0.799$, outperforming unweighted closed-form approximations (0.706; nominal $p = 0.019$). Declared QoS contracts provide measurable signal (+0.095). When publication rates are directly incorporated into an analytical expansion ($\hat{I}_{\text{dyn},1}^{\text{rate}}$), it achieves $\rho = 0.830$ without training, demonstrating that rate transport physics drives the unweighted gap, while the surrogate remains justified when non-linear buffer saturation

must be learned. The GNN trained on the same labels does not (0.598). This study does not show that either oracle, or any of their proxies, can predict real outages.

Choosing a ranker. Table 11 summarizes the evidence by oracle for predictors only. Practitioners must interpret these recommendations as conditional on simulation fidelity. If the discrete-event or cascade simulator does not capture specific physical failure modes (e.g., partial network partitions or disk thrashing), ranking efficacy on physical clusters could diverge. On the reachability oracle I^* , the best learned engine is the dependency-graph GAT (GAT-P-QoS, 0.748), followed by the hybrids (0.657–0.683); all require training, and none is statistically distinguishable from the direct-dependent reference (0.764). Where the reachability ranking itself is wanted, running I^* directly is the simplest and cheapest option at corpus size. Where the queue-flow notion of impact matters (I_{dyn}), a learned surrogate trained on queue-flow labels (GBM-P-QoS→dyn, 0.799) is the only learned ranker that outperforms unweighted closed-form approximations, and the only place where declared QoS contracts carry measurable signal, while a rate-weighted expansion reaches 0.830 without training. Where failure involves fragmentation and throughput loss (I_{comp}), training-free topological scores (Topo-QoS, $\rho = 0.702$) rank highest because their betweenness and articulation terms directly align with fragmentation; among learned models, pure neural rankers collapse (≤ 0.334), while hybrids retain robustness ($\rho = 0.585$). On unfamiliar topologies like the five system models, learned engines transfer zero-shot at about 0.81 against 0.526 for the training-free baseline. Still, active-stratum agreement is weak ($\rho_{>0} \leq 0.342$), and the result rests on five models by one author.

Mitigating Alert Fatigue with a Two-Tiered Triage Protocol. Using a top-20% threshold on GAT-P-QoS recovers about half of the true top-20% set on I^* (0.48), while the rest of the flagged items are not in that set. Figure 5 shows the trade-off: flagging the top 40% by GAT-P-QoS recovers 80% of the critical set (0.81), but Topo-QoS does not reach this level within the top 50%, and no predictor reaches 90% within the top 50%. Reference methods do not perform better (InDeg needs the top 45%). To catch most critical components, a gate would need to flag nearly half the system, making these rankers unsuitable as automated pass/fail filters. In a system with over 100 services, flagging 40–50 would overwhelm developers with alerts.

To operationalize these models constructively, we recommend a *two-tiered triage protocol* rather than an automated blocking gate:

1. **Tier 1 (Commit-stage triage):** When reviewing pull requests, run a quick static dependency count (about 11 ms, around 0.086 Wh at most). Show the top 5 impacted components as an informational note in the pull request review. This highlights potential cascade risks for peer review without blocking developers or causing alert fatigue.
2. **Tier 2 (Staging and sprint verification):** During nightly builds or sprint architecture reviews, use the learned dynamic surrogate (GBM-P-QoS→dyn) to check for nonlinear queue saturation under simulated load. Set a fixed inspection budget, such as the top 10 highest-ranked services, for deliberate resilience improvements, redundant broker routing, or chaos fault injection.

Cost and Green CI/CD. As shown in Table 10, the count is the cheapest ranker, followed by a single pass of I^* . Feature extraction for learned models is the most expensive, costing 4.5 to 72.5 times as much as a single pass on the corpus. If you need reachability ranking, running I^* is simpler and cheaper than using an approximation. Learning is only worthwhile when the oracle is expensive, such as with the queue-flow simulator. From a Green CI/CD and software sustainability perspective [23, 22, 21], adding impact ranking to deployment workflows creates an energy trade-off: running 12.7 CPU-hours of dynamic queue-flow simulation for every commit uses up to about 355.6 Wh (about 1.28 MJ at 28.0W TDP) for the whole corpus, while static dependency counting uses only about 0.086 Wh (310 J; see Section 6.4). For commit-stage triage, training-free structural metrics save considerable energy and carbon, while learned surrogates (GBM-P-QoS→dyn) spread out the simulation cost for later staging checks.

7.2. When Graph Learning Helps, and When It Does Not

Architectural Decision Framework across Operational Regimes. To help with architectural choices, the results group into five main operational regimes based on accuracy levels and architectural types (see replication artifact `results/engine_regimes.json` for details):

Table 11: Which predictor the evidence supports, by the failure notion the oracle encodes (registered secondary and exploratory; Section 6; conditional on simulation fidelity). Reference rankings are excluded because they restate I^* ’s propagation rule (Section 4.4).

Failure notion (oracle)	Ranker	Evidence
Reachability cascade (I^*)	Run I^* directly	0.01–0.72 s per pass; cheaper than any learned engine’s features; best predictor GAT-P-QoS $\rho = 0.748$, indistinguishable from, not equivalent to, the direct-dependent count
Queue-flow delivery loss (I_{dyn})	Dynamic surrogate (GBM-P-QoS → dyn)	$\rho = 0.799$ vs. unweighted first-order expansion 0.706 (Holm $p = 0.019$); amortizes 12.7 CPU-hours (~ 355.6 Wh theoretical TDP proxy) of simulation for staging; rate-weighted closed form reaches 0.830 without training; GNN surrogate reaches 0.598
Fragmentation and throughput (I_{comp})	Topo-QoS (training-free)	$\rho = 0.702$ (betweenness and articulation terms directly align with fragmentation); among learned models, only hybrids retain signal ($\rho = 0.585$), while pure learners collapse (≤ 0.334)
Unfamiliar topology	GAT-P-QoS or GAT-QoS	$\rho = 0.806 / 0.805$ vs. baseline 0.526; $\rho_{>0} \leq 0.342$; five single-author models

1. *Closed-form ranks poorly (Microservices, ATM, IoT Smart City, Healthcare)*: Where the training-free baseline struggles (**Topo-QoS** $\rho = 0.324$), dependency-graph neural models deliver substantial gains (**GAT-P-QoS** $\rho = 0.716$; raw GAT 0.626, HGT 0.604, both outperforming the baseline on 4/4 folds). Forcing a weak baseline prior attenuates performance (hybrids reach 0.502–0.554).
2. *Intermediate regime (ESB, Telecom RAN, Financial Trading, Industrial SCADA)*: In moderately complex topologies where **Topo-QoS** achieves moderate correlation (0.561), hybrid rankers excel (**Hybrid-GAT** $\rho = 0.697$, **Hybrid-HGT** $\rho = 0.681$, winning 4/4 folds). At the same time, **GAT-P-QoS** reaches 0.760.
3. *Closed-form ranks well (Logistics Fleet, AV System, Enterprise, Real-Time Gaming)*: When topological structure corresponds to betweenness (**Topo-QoS** $\rho = 0.775$), pure learned models suffer from overfitting (**GAT-QoS** 0.620, 0/4 folds; **HGT-QoS** 0.672, 1/4 folds). Here, hybrid rankers protect accuracy (**Hybrid-GAT** 0.798, **Hybrid-HGT** 0.786), while **InDeg** reaches 0.858 and **GAT-P-QoS** reaches 0.768.
4. *Unseen, originally pub-sub architectures (Autoware, EdgeX, Home Assistant)*: Learned models generalize zero-shot ($\rho = 0.716$ –0.927) where closed-form baselines fail (0.289–0.534), maintaining positive active-stratum ordering ($\rho_{>0} = 0.183$ –0.833) where baseline correlations turn negative.
5. *Unseen, originally RPC architectures (Online Boutique, Train-Ticket models)*: High full-population correlation ($\rho = 0.710$ –0.810) is driven by separating inert components ($I^* = 0$), but active-stratum ranking remains weak ($\rho_{>0} = -0.19$ to $+0.16$).

Causal Demystification of Message Passing and Inductive Biases. On the raw multigraph, learning mainly used node properties. Three checks explain how these engines worked. First, gradient-boosted decision trees using only per-node features (no graph message passing, **GBM-Feat**) reach $\rho = 0.642$ under LOSO, similar to **GAT-QoS** (0.635). Second, in the raw multigraph, all structural links point away from Applications, so standard forward GAT convolutions do not send messages to the scored nodes—removing all edges does not change their outputs. Third, HGT reaches Applications only through reverse edge passes, and removing them has no effect (**HGT-QoS-U**, -0.010 , $p = 0.91$). So, on the raw multigraph, neural models acted as non-linear combiners of precomputed centralities and degree metrics, not as relational message-passing engines. On the derived dependency graph, directed edges go from dependents to dependencies: three layers of message passing gather information from the true dependent neighborhood, letting **GAT-P-QoS** outperform **GBM-Feat** by $+0.106$. Since every learner also gets an in-degree feature (Section 3.5), this margin does not mean attention finds the dependent count on its own, and ablation tests show it does not: without the degree columns, **GAT-P-QoS** loses 0.136, while a sum-aggregation network of the same size loses 0.011 (Section 6.2).

Homogeneous versus heterogeneous architectures. On the raw multigraph, where message passing into Applications was inactive, adding typing made no measurable difference (main effect -0.014 , interaction

+0.001). On the dependency graph, where message passing is active, the heterogeneous transformer (HGT-P-QoS) was unstable across seeds in the tested configuration (width 100, fixed setup; mean within-fold seed standard deviation 0.254, mean $\rho = 0.514$), while the homogeneous GAT-P-QoS was stable (0.030, $\rho = 0.748$). This means that, in this setup, relation-specific parameters did not improve accuracy, though they reduced stability. This does not mean typing is always harmful. Only HGT-QoS was run under the registered selection rule (Section 6.2); averaging five seeds’ predictions raises HGT-P-QoS to 0.618 (still below the references), and denser topologies or architectures with messages reaching scored nodes may behave differently.

Why the tree surrogate learns queue-flow impact and the GNN surrogate does not. On I_{dyn} , the gradient-boosted surrogate (GBM-P-QoS $\rightarrow$ dyn) achieves $\rho = 0.799$ on held-out architectures, +0.094 above the unweighted first-order expansion $\hat{I}_1^*$ (0.706). The attention surrogate (GAT-P-QoS $\rightarrow$ dyn) reaches 0.598, 0.201 below it. Once trained on I^* labels instead, the same GAT scores within 0.008 of that (control F5), so it does not extract queue-flow signal from its labels. The study does not explain why. The two surrogates differ in both their inputs and model type: the tree model uses the first-order expansion and each Application’s publish rate and payload size, which I_{dyn} reads directly. At the same time, the dependency-graph GAT’s node properties do not include these. Furthermore, the unweighted baseline $\hat{I}_1^*$ (Eq. (6)) was derived for reachability without topic publication rates. When a rate-weighted first-order expansion is evaluated instead— $\hat{I}_{\text{dyn},1}^{\text{rate}}(v) = \sum_{t \in \text{pub}(v)} (r_t / |\text{pub}(t)|) \cdot |\text{sub}(t)|$, where r_t is the declared publication frequency—it achieves a mean $\rho = 0.830$ across the twelve scenarios. This demonstrates that the tree surrogate’s +0.094 advantage over closed-form approximations stems primarily from access to declared rate features rather than tree learning inductive biases alone: when closed-form approximations reflect the rate transport physics of the dynamic queue-flow oracle, they match or exceed the learned surrogate without training. A controlled experiment where a GNN receives the same rate, payload, and expansion features is needed to separate model effects from feature differences.

7.3. Threats to Validity

Construct validity and oracle circularity. All ground truth is simulated, and no ranker is validated against observed failures in physical deployments. We conducted no empirical fault injection (e.g., chaotic network partitions, process crashes, or disk exhaustion) on running distributed clusters. Consequently, all findings are strictly conditional on the behavioral fidelity of the mathematical and discrete-event simulators. Discrete-event simulations omit complex real-world distributed runtime dynamics: kernel TCP buffer bloat, socket starvation, thread pool exhaustion, garbage collection freezes, asymmetric network partition drops, and cascading retry storms. The circularity between the primary oracle and the rankers is analyzed in Section 4.4: the dependency counts are references for I^* , closed-form centrality is proximate to I_{comp} , and the learned engines inherit part of the overlap through their degree features, which ablation experiments show they depend on. Inert components are separated by construction, which is why $\rho_{>0}$ is reported beside ρ . We decided to reclassify the counts as references after all results were finalized, and this change did not affect any numbers. The two other oracles bound, but do not remove, the threat: I_{dyn} shares the topology and the first-order effect, and I_{comp} ’s weights encode a declared rather than an elicited priority vector. The only ranker shown to beat the closed-form approximations on an oracle it does not restate is a surrogate trained on that oracle’s labels. The evaluation is confined to Applications, so dependency Rules 2–4 and 6, which govern brokers and hosts, are untested.

Internal validity, reproducibility, and drift. Predictors consume G_{analysis} while simulation oracles execute on $G_{\text{structural}}$, verified by regression tests. We held model capacity, depth, and early stopping constant across comparison arms. The registered selection rule was not applied to the published learned results, a deviation disclosed in Section 5.3; a post hoc sensitivity check applies it to HGT-QoS and GAT-P-QoS (Section 6.2), and every other statement regarding a learned engine is conditional on one fixed configuration. The learned engines are also seed-sensitive. Across the five seeds of a fold, the mean within-fold standard deviation of ρ is 0.099 for HGT-QoS, 0.254 for HGT-P-QoS, 0.024 for GAT-QoS and 0.030 for GAT-P-QoS, and single HGT seeds differ from others in the same fold by more than 1.0. Averaging the five seeds’ predictions instead of their ρ values raises every learned engine, most of all the unstable ones (HGT-P-QoS from 0.514 to

0.618); the tables report the registered statistic, the mean of per-seed ρ . Across revision cycles and compute devices, learned cells drifted by up to 0.172 (± 0.041 on HGT-QoS), more than several effects this paper reports (the primary contrast, +0.069; the QoS main effect, +0.073). For that reason, every learned cell in Tables 5, 7 and 9 is reported from one named CPU sweep whose per-seed logs reproduce the published mean exactly, the sensitivity verification sweep re-ran every learned comparator to an exact match, and a hierarchical bootstrap that resamples seeds within folds widens the learned rows’ intervals only slightly. The Connectivity Degradation Index samples nodes with hash-dependent tie-breaking, so PYTHONHASHSEED=0 is fixed. We reconcile all reported values mechanically against released artifacts.

External validity and the single-modeller threat. Synthetic topologies come from a single generator family, which may induce parametric regularities not representative of heterogeneous industrial architectures. The five system models are hand-authored approximations (22–41 Applications) of open-source projects, each written by the first author from public documentation. All five RQ3 results are exposed to this threat: no second modeler authored any model independently, and no inter-modeler reliability metric is reported. To establish a baseline for future multi-rater replication, we conducted an exploratory calibration spot-check on the two smallest systems (Home Assistant and EdgeX Foundry), evaluating node-level Jaccard overlap and edge concordance between the primary model and a structured re-extraction by the second author. While service entity identification showed high agreement ($J \geq 0.88$), edge multiplicity and topic topology diverged ($J \approx 0.65\text{--}0.72$), showing that broker channel synthesis from documentation requires non-trivial modeler interpretation. The system models also differ structurally from the synthetic folds: a larger share of their Applications has zero simulated impact, and their fan-in is more concentrated, favoring **Reach** and inflating full-population ρ . Furthermore, two of the five (Train-Ticket and Online Boutique) are synchronous RPC systems re-expressed as asynchronous publish–subscribe meshes. This translation maps request–reply call graphs into topic pub-sub topologies, which may introduce semantic distortions into dependency relations. We did not evaluate architectures with tens of thousands of components; we timed the counting path up to 10,000 components (Section 6.4), and we did not evaluate the learned engines or the oracles. We did not reproduce published learned criticality models (FINDER, DrBC) because they do not support typed multigraphs or pub-sub semantics.

Sustainability and energy estimation validity. Energy metrics reported for continuous integration workflows (Section 6.4) represent theoretical upper bounds derived from processor thermal design power (28.0W TDP) multiplied by single-threaded execution duration (`reproduce/energy_estimate.py`). They do not capture hardware-level power variations such as dynamic voltage and frequency scaling (DVFS), DRAM or disk energy, or multi-socket server idle power, and they were not verified with physical hardware meters (e.g., RAPL or external power analyzers). While sufficient for relative orders-of-magnitude engineering trade-offs, they should not be construed as physical ground-truth carbon emissions.

Conclusion validity. The only confirmatory results are the plan’s two contrasts, the null primary and the null secondary (Section 5.3). Everything else, including the two hybrid contrasts, is registered secondary or exploratory and was designed after the primary null was known. LOSO folds share ten of eleven training scenarios, so the Wilcoxon p -values are nominal and anti-conservative; the Nadeau–Bengio corrected test, which accounts for the overlap, leaves the hybrid contrasts significant and the primary null. For training-free rankers, the folds are simply twelve scenarios from one generator, so the effective sample is small, and with twelve folds a bootstrap CI over folds is coarse; over five system models it is indicative only. Equivalence, which a claim that one ranker “matches” another requires, is not established for any pair at ± 0.05 .

7.4. Limitations and Future Work

The study acknowledges several limitations: (1) it has not been validated against real outages or post-mortems on physical testbeds with chaotic fault injection (such as Chaos Mesh or ChaosBlade), though reference systems with known faults, like Train-Ticket [27], and MQTT-based EdgeX and Home Assistant deployments are good first targets; (2) there is no controlled comparison with strict feature parity between GNNs and tree ensembles for dynamic queue-flow prediction; (3) a full hyperparameter search was not done for every learned model, only a stage-1 selection for two engines; (4) there is no independently authored multi-rater replication of each open-source system, nor formal inter-modeller agreement metrics (like Jaccard edge concordance); (5) automatic extraction of architecture graphs from ROS 2 launch files, Kubernetes

manifests, and Helm charts is not addressed; (6) empirical, hardware-based energy and carbon accounting (such as via RAPL) in CI pipelines was not performed; (7) synchronous call edges are not modeled by the dependency rules; and (8) the proposed explanation layer has not been validated. An appropriate next step is to build a GNN surrogate for the queue-flow oracle that uses sum aggregation and receives the same QoS and expansion features, to see if neural architectures can match tree surrogates for dynamic queue-flow dynamics.

8. Conclusion

Software-as-a-Graph (SaG) derives explicit dependency graphs from publish-subscribe deployment manifests and uses them to rank components by simulated cascade impact before deployment. We evaluated graph neural networks on the raw multigraph and on the dependency graph, as well as hybrid engines and gradient-boosted surrogates. Each was compared against a training-free baseline and against reference rankings that restate each simulator’s rule. The evaluation spans twelve synthetic architectures, five hand-crafted models of open-source systems, and three failure simulators. Five conclusions follow.

1. **Explicit dependency graphs are what make graph learning effective.** In the raw multigraph, messages never reach the scored Applications, and the registered primary contrast showed no significant difference relative to the training-free baseline ($+0.069$, $p = 0.266$). On the derived dependency graph, the same learners improve by $+0.08$ to $+0.11$. A graph attention network reaches $\rho = 0.748$ under leave-one-scenario-out cross-validation, and about 0.81 zero-shot on the system models.
2. **A classical coupling metric captures most of the reachability impact.** Counting direct dependents, i.e., publish-subscribe afferent coupling, yields $\rho = 0.764$. This is statistically indistinguishable from the best learner, which approaches it by drawing on its degree features. The reference criterion makes this visible: because the count restates the oracle’s first propagation wave, its strength measures how much of the oracle is circular rather than predictive skill. In practice, it is a training-free triage signal that costs one to two orders of magnitude less than a single simulation pass.
3. **Learned surrogates pay off where simulation is expensive.** On a queue-flow oracle that takes 12.7 CPU-hours to label the corpus, a gradient-boosted surrogate combining dependency, rate, payload and QoS features ranks held-out architectures at $\rho = 0.799$, outperforming unweighted closed-form approximations (0.706; nominal $p = 0.019$) and replacing hours of simulation with inference in seconds. Declared QoS contracts contribute measurable signal ($+0.095$). However, when declared message rates are incorporated into an analytical expansion ($\hat{I}_{\text{dyn},1}^{\text{rate}}$), it achieves $\rho = 0.830$ without training, demonstrating that the surrogate’s advantage over the unweighted baseline stems from rate transport physics rather than machine learning inductive biases alone. Learning remains valuable where closed forms cannot capture complex buffer saturation.
4. **Anchoring and transfer trade-off.** On a multi-criteria oracle combining fragmentation and throughput loss, hybrid engines that correct a closed-form prior remain accurate ($\rho = 0.585$), whereas learners without a prior degrade (≤ 0.334). Training-free scores aligned with that oracle’s terms rank higher still. Pure learned engines transfer better to unfamiliar architectures (0.76–0.81 vs. 0.66–0.70), mainly by separating components that propagate failures from those that do not. Ordering among active components ($\rho_{>0} \leq 0.342$) remains the main open challenge.
5. **The evidence supports an energy-aware two-tier triage process.** Recovering 80% of the true critical set requires reviewing roughly the top 40% of Applications, so these rankings are best used to prioritize architectural review rather than to block commits. Feature extraction costs $4.5\text{--}72.5\times$ one reachability pass. Dependency counting therefore suits commit-stage triage. Learned surrogates suit staging, where they replace an expensive simulation, at an estimated fraction of its theoretical energy cost (~ 355.6 Wh proxy).

These conclusions are relative to simulated rather than observed failures, and the system models were authored by a single modeler. Three extensions would test how far the results reach. First, validating rankings against real outages or controlled fault injection on reference systems such as Train-Ticket or MQTT-based

deployments. Second, extracting architecture graphs automatically from ROS 2 launch files and Kubernetes manifests. Third, giving graph neural network surrogates the same rate, payload and expansion features as the tree model, to determine whether the queue-flow advantage belongs to the features or to the model family. Because the synthetic corpus regenerates byte-identically and every reported figure is reconciled against released artifacts, others can extend or challenge these results directly.

Declarations

CRedit Authorship Contribution Statement. **Ibrahim Onuralp Yigit:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Feza Buzluca:** Conceptualization, Methodology, Validation, Writing – review and editing, Supervision.

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Data Availability. The replication package (datasets, harnesses, checkpoints, scripts) is available on Zenodo under DOI 10.5281/zenodo.14922108 [75] with `uv/pip` environments. The public repository documents every experiment reported here — protocol, hyperparameters, reproduction command, artifacts and extended results — at <https://github.com/onuralpyigit/software-as-a-graph/tree/main/docs/research/jss/experiments>. Synthetic datasets regenerate byte-identically. The deposit ships all artifacts backing reported tables as a dated bundle (`SaG_JSS_Results_<stamp>`) with a `MANIFEST.json` recording SHA-256 digests, commit hashes, and corpus provenance. Four archived artifacts predate provenance stamping and carry no commit or corpus digest: `atm_scale_sweep_v3.json`, `qos_label_ablation.json` (Section 4.3), `threshold_sensitivity_v3.json` and `topic_weight_sensitivity_v3.json`; their correspondence to the corpus is asserted by the bundle rather than recorded in the file. In the main text, only the two label-ablation correlations of Section 4.3 ($\rho = 0.965$ and 0.977) rest on one of them; the other three back extended sensitivity sweeps in the repository only, and no number in the abstract or in Tables 5–9 depends on any of the four. The verification script (`reproduce/reconcile_manuscript.py`) runs standalone against the deposit, mechanically verifying 1,548 reported figures against the JSON artifacts.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work, the authors used Anthropic’s Claude Opus 5 and Google’s Gemini Flash 3.8 for drafting manuscript revisions, LaTeX typesetting and formatting assistance, and developing analysis scripts in the replication package, and Grammarly for grammar checking and language copy-editing. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article. The study design, the choice of experiments, the interpretation of results, and all scientific claims are the authors’ own.

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